



Failure zone homogenization for modeling damage- and debonding-induced softening in composites including gradient-extended damage at finite strains

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ABSTRACT

Within multiscale modeling strategies for composite materials, standard Hill averaging is widely used for homogenization of repeating unit cells (RUCs). However, Hill's homogenization approach reaches its limitations at the presence of strain softening, because in the standard averaging sense the representativeness of the considered micro-scale volume is lost. Hence, in order to overcome these limitations, we propose a new failure zone averaging scheme at finite strains, which is also applicable in the softening regime. First, starting from Hill's assumption of equal virtual work densities on different scales, a novel homogenization approach is derived analytically. Then, the analytical results are verified by numerical simulations evaluating different RUC sizes and accounting for varying geometric realizations in a statistical manner. It is shown that the proposed homogenization scheme yields size-independent stress-strain relations even after localized damage has occurred leading to strain softening within the RUCs. In addition, the material model for the epoxy bulk is valid for the description of damage progression at large deformations using a gradient-enhanced damage model. It turns out that the micromorphic power density – related to the gradient enhancement – of the failure zone gives significant values during the formation of localization while the overall work density is small. Thus, it can be concluded that for accurate failure zone homogenization, the contribution of gradient terms has to be considered. Further, cohesive zone elements are incorporated to consider the decohesion between fibers and matrix. As a result, the stress-strain curves show a lower peak stress and a higher scatter in the stress response compared to samples with perfect interfaces. In contrast, the work density plots show a higher dissipation for the RUC with damaging interfaces.

1. Introduction

Due to high stiffness and strength combined with low density, carbon fiber reinforced plastics (CFRPs) were used in a growing number of applications (see, e.g. Rösler et al., 2013). For example, in civil engineering, new construction methods used CFRPs as reinforcement of concrete as was shown, e.g. in Karayannis et al. (2018). Additionally, the beneficial properties of CFRPs had been exploited in applications such as aerospace technology or wind turbines for a long time. Most commonly, CFRPs were fabricated either as laminates with tailored material properties or with a woven structure on the micro level. In the latter, even simple loading conditions such as uniaxial tension led to inhomogeneous strains and complex load states within the microstructure due to the complex microscopic geometry (see, e.g. Stier et al., 2015 or Höwer et al., 2019).

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Since CFRPs with a woven microstructure generally showed a complex damage behavior, it is hardly possible to describe this behavior with only one scalar damage variable and one damage function. As was shown in e.g. [Meer and Sluys \(2009\)](#), [Mukhopadhyay and Hallett \(2019\)](#) and [Poggenpohl et al. \(2020\)](#) or [Dean et al. \(2021\)](#) one approach was to use a set of scalar damage variables to differentiate between damage of the fibers and damage of the matrix. Alternatively, one damage variable with two different damage functions for fiber and matrix as shown, e.g. in [Mandal et al. \(2020\)](#) was used. In other models, six scalars (see, e.g. [Chen and Aliabadi, 2019](#) or [He et al., 2020](#)) or even a fourth order Tensor (see, e.g. [Chow and Wei, 1999](#) or [Ryvkin and Aboudi, 2020](#)) was used to account for anisotropic stiffness degradation of CFRPs. Another approach was shown, e.g. in [Simon et al. \(2017\)](#) and [Bednarczyk et al. \(2015\)](#), where a general framework for damage evolution in woven composites was given. There, a damage interaction matrix was introduced to control the evolution and interaction of different damage mechanisms. However, the damage evolution and interaction of those models have to be justified numerically or experimentally. Here, RUC simulations offer the opportunity to grant insight into the damage evolution since the loading scenario can be controlled precisely.

Additionally, CFRPs generally fail in a brittle manner. Thus, high safety factors are required in order to prevent catastrophic failure. A more accurate failure prediction is desired, since it leads to decreasing factors of safety and consequently better material performance (see, e.g. [Wu et al., 2020](#)). More insight into the microscopic failure mechanisms of CFRPs is needed.

The analysis of representative unit cells (RUCs) in a staggered or FE^2 scheme is a helpful method to gain a better understanding of the mechanisms at the micro level. For that, Hill's approach (see [Hill, 1967, 1972](#)) is widely used to compute a homogenized mechanical as well as thermal responses from the results obtained from RUC simulations (see, e.g. [Mikeš et al., 2021](#) for a comparative study). Hill's approach is used exemplary in [Tsalis et al. \(2013\)](#) for elastoplastic composites, in [Chatzigeorgiou et al. \(2016\)](#) for a thermomechanically coupled material, in [Meade et al. \(2021\)](#) to predict the ductile failure of high strength aluminum due to void growth or in [Cao et al. \(2020\)](#) for a FFT based analysis of concrete cracking just to name a few. Additionally, as suggested, e.g. in [Settgast et al. \(2020\)](#), Hill's approach is used to generate training data for neuronal network applications. As can be seen, e.g. in [Elnekhaaily and Talreja \(2018\)](#) and [He et al. \(2020\)](#) Hill's approach was used as well to analyze the effect of micro geometry on the homogenized behavior of carbon fiber reinforced materials. In [Elnekhaaily and Talreja \(2018\)](#), RUC simulations were used to investigate the influence of fiber distribution on the failure onset. However, the simulation was restricted to the onset of failure, while the failure process itself was not analyzed. In [He et al. \(2020\)](#), a multiscale simulation using several scales was presented. The analysis took into account the micro scale which consisted of carbon fibers and epoxy matrix, the meso scale which consisted of the woven yarns and the macro scale on the structural level. The material model (invoking elasticity, plasticity and failure as well as temperature dependency of the material parameters) was tested against experimental data. However, the RUC size was only determined by the means of the elastic response as was suggested in [Melro et al. \(2012\)](#). As will be shown later, the definition of a representative RUC size might be different in the hardening and softening regime, where for the latter, the convergence should be investigated individually.

It was shown in several publications, for instance in [Gitman et al. \(2004, 2007\)](#) or [Geers et al. \(2010\)](#) that standard homogenization using Hill's approach was not applicable at the presence of localization phenomena and a representative size of the RUC could not be uniquely defined. In [Gitman et al. \(2007\)](#), it was concluded that a mismatch of dimensions between dissipative volume and RUC volume (i.e. the volume of the stored elastic energy) was the main cause that Hill's approach was not applicable.

In order to counteract this inconsistency, the failure zone averaging method was developed in [Nguyen et al. \(2010\)](#). There, the failure zone averaging method was shown to be insensitive to the size converging problem of Hill's approach for RUCs undergoing localizing phenomena. This method circumvented the mismatch between elastic energy and dissipated energy, since the analyzed domain was restricted to the actively damaging volume of the RUC. Using this method, a representative size of the RUC was identified. In more recent publications, computational aspects were investigated ([Nguyen et al., 2011](#)) and the method was applied to a FE^2 scheme with complex microstructure of concrete aggregates at the micro level ([Nguyen et al., 2012](#)). More recently, the failure zone averaging method has been applied to microscopical cracks using cohesive zone elements (see, e.g. [Turteltaub et al., 2018](#); [Turteltaub and de Jong, 2019](#); [Turteltaub and Suárez-Millán, 2020](#) or [Ke and Van Der Meer, 2021](#)). There, cohesive zone elements were used at the micro scale to model the debonding of the interface as well as cracks of the bulk material.

In this publication, the failure zone averaging method is adapted to the investigation of repeating unit cells (RUCs) of long carbon fiber reinforced plastics subjected to transversal tension. For the failure of the resin matrix, an isotropic brittle damage model at finite strains is used while the fibers behave linear elastically at finite strains. To overcome undesired mesh size sensitivity, the micromorphic approach as introduced in [Forest \(2009\)](#) is adopted, leading to a gradient-enhancement in the damage formulation. As shown e.g. in [Peerlings et al. \(1996\)](#) or [Peerlings et al. \(2001\)](#), the introduction of higher order gradients can cure undesired mesh size sensitivity. In recent publications, gradient enhanced damage is applied to damage-plasticity (see, e.g. [Miehe et al., 2016](#)), damage-viscoplasticity (see, e.g. [Mozaffari and Voyiadjis, 2016](#)) as well as thermo-mechanically coupled damage-plasticity (see, e.g. [Felder et al., 2022](#)).

2. Material modeling

In the following, the material models for the bulk (which in turn consist of epoxy matrix and carbon fibers) and interfaces are presented. The nomenclature of vectors, tensors and base operators – which are defined exclusively in the euclidean space – is summarized in the following:

a, A	Scalar	
$\mathbf{a}$	First order tensor	
$\mathbf{a}$	Vector $n \times 1$	
$\mathbf{A}$	Second order tensor	
$\mathbf{A}$	Matrix $n \times m$	
$\mathbf{A}$	Fourth order tensor	
$\cdot$	Single contraction, i.e.	$\mathbf{A} \cdot \mathbf{B} = A_{ij} B_{jk}$
$:$	Double contraction, i.e.	$\mathbf{A} : \mathbf{B} = A_{ij} B_{ij}$
$\mathbf{a} \otimes \mathbf{b}$	Dyadic product, i.e.	$\mathbf{a} \otimes \mathbf{b} = a_i b_j$
$\mathbf{A}^T$	Transpose, i.e.	$A_{ij}^T = A_{ji}$

2.1. Bulk material model

For the simulation of the bulk material behavior of the RUC, the isotropic part of the material model introduced in Poggenpohl et al. (2020) was used. In the following, the material model is introduced and a brief summary of the implementation into a finite element code is presented. For more details, the reader is referred to the publication mentioned above.

The material model presented in the following is of brittle damaging type. As shown, e.g. in Rocha et al. (2019) or Morelle et al. (2017), epoxy can show a significant amount of plasticity until failure occurs. However, plastic strains are often dependent on triaxial stress states. From the experimental data given in Poggenpohl et al. (2020), which showed no considerable amount of irreversible deformation, a brittle damaging behavior of the epoxy matrix was deduced.

Overall, the volume specific free energy of the material consist of four parts:

$$\psi(\mathbf{E}, D, D^\chi) = \psi_{\text{mech}}(\mathbf{E}, D) + \psi_h(D) + \psi_g(\nabla_0 D^\chi) + \psi_\chi(D, D^\chi) \quad (1)$$

In above Eq. (1), the scalar damage variable D describes the amount of stiffness degradation of a material point. Here, $D=0$ refers to a virgin material with no stiffness degradation, while $D=1$ refers to a complete loss of material stiffness. Generally, $0 \leq D \leq 1$ holds in case of numeric simulations. The variable $\mathbf{E}$ is the Green–Lagrange strain tensor which is defined as

$$\mathbf{E} = \frac{1}{2}[\mathbf{F}^T \cdot \mathbf{F} - \mathbf{I}], \quad (2)$$

where $\mathbf{I}$ is the unit tensor and $\mathbf{F}$ is the deformation gradient defined as the derivative of the current (deformed) configuration $\mathbf{x}$ with respect to the reference (undeformed) configuration $\mathbf{X}$:

$$\mathbf{F} = \nabla_0 \mathbf{x} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}} \quad (3)$$

As can be seen in Eq. (3), ∇_0 refers to the spatial gradient with respect to the reference configuration. Furthermore, the variable D^χ in Eq. (1) is the micromorphic counterpart of the local damage variable D and is treated as an additional degree of freedom. The four parts of the free energy density (1) are discussed in the following.

The material model used for the elastic response (ψ_e) is the St. Venant–Kirchhoff material model which in general reads

$$\psi_e^*(\mathbf{E}) = \frac{\lambda}{2} \text{tr}(\mathbf{E})^2 + \mu \text{tr}(\mathbf{E}^2). \quad (4)$$

Here, parameters λ and μ are the Lamé constants and $\text{tr}(\mathbf{A})$ denotes the trace of Tensor $\mathbf{A}$, i.e. $\text{tr}(\mathbf{A}) = A_{ii}$. Inspired by Fassin et al. (2019b), a tension–compression asymmetry is introduced in the stiffness degradation due to local material softening in the first term of Eq. (4) giving the mechanical part of the total free energy:

$$\psi_{\text{mech}}(\mathbf{E}, D) = [1 - D]^n \frac{\lambda}{2} [\text{tr}^+(\mathbf{E})]^2 + (1 - h_{tc} D)^n \frac{\lambda}{2} [\text{tr}^-(\mathbf{E})]^2 + [1 - D]^n \mu \text{tr}(\mathbf{E}^2) \quad (5)$$

In the above, for $n = 1$ the present model reduces to the equivalent strain model, where for $n = 2$ the energy equivalent model is obtained. In addition, the degree of crack closure is controlled through the parameter h_{tc} . For $h_{tc} = 1$ damage influences the compression part in the same way as the tension part, no crack closure effects are present. Complementary, for $h_{tc} = 0$ damage is not considered in the compressive regime at all, which corresponds to the highest degree of crack closure. Generally, intermediate states (i.e. $0 \leq h_{tc} \leq 1$) are used. The tension ($\text{tr}^+(\mathbf{E})$) and compression ($\text{tr}^-(\mathbf{E})$) parts are defined as

$$\text{tr}^+(\mathbf{E}) = \langle \text{tr}(\mathbf{E}) \rangle \quad \text{and} \quad \text{tr}^-(\mathbf{E}) = -\langle -\text{tr}(\mathbf{E}) \rangle. \quad (6)$$

Here, $\langle a \rangle$ denotes the Macauley brackets:

$$\langle a \rangle = \begin{cases} 0 & \text{for } a \leq 0 \\ a & \text{for } a > 0 \end{cases} \quad (7)$$

Additional to the mechanical part, damage hardening is introduced, giving the second part of the free energy

$$\psi_h = \frac{K}{2} D^2 \quad (8)$$

In the equation above, parameter K controls the degree of damage hardening. It shall be noted that the terminology originates from the analogy to plastic hardening. In Fig. 1, a Gauss point study of parameter K is shown. The material parameters used are those

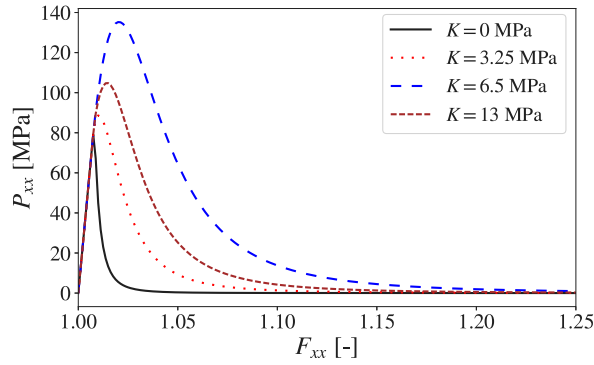


Fig. 1. Stress-stretch curves for different hardening parameters K .

for the matrix material shown in Table 1 in Section 6. As can be seen, $K = 0$ MPa leads to a very brittle material response while for increasing K , damage is delayed. Higher values of K lead to a slower evolution of damage and thus, a slower decrease of strength and stiffness as well as an increase of maximum load.

When using material models which exhibit softening, strain localizations may occur, which causes undesired mesh dependency of the solution. Therefore, the micromorphic extension as introduced, e.g. in Forest (2009) is applied. Micromorphic energetic terms ψ_g and ψ_χ (the last two parts of the free energy) are defined as

$$\psi_\chi = \frac{H^\chi}{2} [D^\chi - D]^2 \quad (9)$$

$$\psi_g = \frac{El^2}{2} [\nabla_0 D^\chi \cdot \nabla_0 D^\chi]. \quad (10)$$

In Eqs. (9) and (10), the micromorphic extension consists of a coupling term ψ_χ and a gradient term ψ_g . The coupling term of Eq. (9) with penalty factor H^χ couples the local and the micromorphic damage variable and is chosen sufficiently high, such that the difference between D^χ and D is at least two magnitudes smaller than the numeric value of D itself. As suggested in Fassin et al. (2019a), the Young's modulus E and an internal length scale factor l are used within ψ_g .

Due to the micromorphic extension of the local energy, an extended Clausius–Duhem inequality (see, e.g. Forest, 2019) is obtained

$$-\dot{\psi} + S : \dot{E} + \underbrace{a \dot{D}^\chi + b \cdot \nabla_0 \dot{D}^\chi}_{\text{micromorphic extension}} \geq 0, \quad (11)$$

where S is the second Piola–Kirchhoff stress tensor, while a and b denote the work conjugate stress-like quantities of the additional degree of freedom D^χ and its gradient $\nabla_0 D^\chi$, respectively. Applying the chain rule on $\dot{\psi}$ and rearranging the terms results in the following expanded inequality:

$$\left[S - \frac{\partial \psi}{\partial E} \right] : \dot{E} + \left[a - \frac{\partial \psi}{\partial D^\chi} \right] \dot{D}^\chi + \left[b - \frac{\partial \psi}{\partial \nabla_0 D^\chi} \right] \cdot \nabla_0 \dot{D}^\chi - \frac{\partial \psi}{\partial D} \dot{D} \geq 0 \quad (12)$$

To ensure that Inequality (12) holds for arbitrary processes, it is chosen, following the argumentations of Coleman and Noll (1961) and Forest (2016), to set the brackets to zero. This results in the following relations:

$$S = \frac{\partial \psi}{\partial E}, \quad a = \frac{\partial \psi}{\partial D^\chi}, \quad b = \frac{\partial \psi}{\partial \nabla_0 D^\chi} \quad (13)$$

The introduction of the stress-like quantities a and b leads to the micromorphic balance equation, which has to be solved additionally to the balance of linear momentum with f_0 denoting the conventional body forces (see, e.g. Peerlings et al., 1996, 2004 or Forest, 2016). Hence, the balance equations, which have to be solved read:

$$\text{Div}(\mathbf{F} \cdot \mathbf{S}) + \mathbf{f}_0 = \mathbf{0}; \quad \text{Div}(\mathbf{b}) - a = 0 \quad \text{in } \Omega_0 \quad (14)$$

Eqs. (14) form the strong form of the boundary value problem which have to be fulfilled in every material point of the bulk of body Ω_0 . Here, $\mathbf{f}_0$ denotes the conventional body forces.

To complete the boundary value problem, Dirichlet and Neumann conditions are defined on the boundary:

BC for balance of linear momentum:	BC for micromorphic balance:
$[\mathbf{F} \cdot \mathbf{S}] \cdot \mathbf{n}_0 = \mathbf{t}_0 \quad \text{on } \partial\Omega_{0,n}$	$\nabla_0 D^\chi \cdot \mathbf{n}_0 = 0 \quad \text{on } \partial\Omega_0$
$\mathbf{u} = \hat{\mathbf{u}} \quad \text{on } \partial\Omega_{0,d}$	

(15)

As can be seen in Eq. (15), the boundary of the body Ω_0 is decomposed into a Neumann ($\partial\Omega_{0,n}$) and a Dirichlet part ($\partial\Omega_{0,d}$) for the balance of linear momentum. However, in case of the micromorphic balance equation, only Neumann-type boundary conditions

are prescribed on $\partial\Omega_0$. Furthermore, the gradient of D^χ has to vanish in normal direction on $\partial\Omega_0$, which is equivalent to requiring the crack (or localization zone) to be perpendicular on the boundary. This might seem as an arbitrary restriction, but also widely used in other publications (see, e.g. [Clasen et al., 2013](#); [Holthusen et al., 2020](#); [Forest, 2009, 2016](#) and [Fassin et al., 2019a; Brepols et al., 2020](#)) and will be of importance in the following Section 4.

After application of Eq. (13) on the Clausius–Duhem inequality (12), the reduced dissipation inequality has to hold:

$$\underbrace{-\frac{\partial\psi}{\partial D}}_{=:Y} \dot{D} \geq 0 \quad (16)$$

Here, the damage driving force Y is introduced. Accordingly, the damage driving force for the material model given reads:

$$Y = -\frac{\partial\psi}{\partial D} = n[1 - D]^{n-1}\psi_e + H^\chi [D^\chi - D] - KD. \quad (17)$$

Here, the coupling between the local damage variable D and the micromorphic damage variable D^χ becomes apparent. Using the damage driving force Y , a loading function analogous to the yield function in plasticity can be constructed, where $Y_0 \geq 0$ indicates the damage threshold variable.

$$\Phi(Y) := Y - Y_0 \leq 0 \quad (18)$$

Loading and unloading conditions (Karush–Khun–Tucker conditions) ensure that the Clausius–Duhem inequality is fulfilled at any time. For the given damage model, they read:

$$\dot{D} \geq 0, \quad \dot{D}\Phi = 0, \quad \Phi \leq 0 \quad (19)$$

Noticeably, damage healing is not considered here.

2.2. Interface material model

For the simulation of the interface response between epoxy matrix and carbon fibers, the cohesive zone (CZ) formulation of [Rezaei et al. \(2017\)](#) is used. Generally speaking, an interface connects two phases $\Omega_{0,1}$ and $\Omega_{0,2}$, while $\mathbf{g} = [g_n, g_{s1}, g_{s2}]^T$ and $\mathbf{t} = [t_n, t_{s1}, t_{s2}]^T$ describe the gap and traction vectors between the two phases. Here, subscripts $n, s1$ and $s2$ denote the components in normal and both shear directions. The free energy per unit area is given by

$$\psi_{CZ} = \frac{1}{2}(1 - D_{CZ})k_0\lambda_{CZ}^2 + \frac{1}{2}k_p\langle -g_n \rangle^2, \quad (20)$$

with $\lambda_{CZ} = \sqrt{\langle g_n \rangle^2 + \beta^2(g_{s1}^2 + g_{s2}^2)}$ denoting the effective separation, where β defines the influence of shear components (see, e.g. [Simon et al., 2015](#)). The initial stiffness of the interface is denoted by k_0 . In the present study, β is set to 1.0 so that the normal and shear gap contribution to the effective separation is the same. Self-penetration of the interface is prevented by the last term with penalty parameter k_p and the negative normal gap $\langle -g_n \rangle$.

From Eq. (20), the components of the stress vector can be derived

$$t_n := \frac{\partial\psi_{CZ}}{\partial g_n} = (1 - D_{CZ})k_0\langle g_n \rangle - k_p\langle -g_n \rangle \quad (21)$$

$$t_{s1} := \frac{\partial\psi_{CZ}}{\partial g_{s1}} = (1 - D_{CZ})k_0 g_{s1}\beta^2 \quad (22)$$

$$t_{s2} := \frac{\partial\psi_{CZ}}{\partial g_{s2}} = (1 - D_{CZ})k_0 g_{s2}\beta^2. \quad (23)$$

In addition, the effective traction t_{CZ} can be derived as well, giving

$$t_{CZ} = \frac{\partial\psi_{CZ}}{\partial\lambda_{CZ}} = \sqrt{t_n^2 + \beta^{-2}(t_{s1}^2 + t_{s2}^2)} = (1 - D_{CZ})k_0\lambda_{CZ}. \quad (24)$$

A bilinear traction-separation law is adopted (see [Fig. 2](#)). It is defined by the maximum traction t_0 , the separation at maximum traction λ_0 and the separation at final failure λ_f . Based on the traction-separation law, the following damage evolution equation has to hold:

$$D_{CZ} = \begin{cases} 0 & \text{if } \lambda < \lambda_0 \\ \frac{\lambda_f}{\lambda_f - \lambda_0} \frac{\lambda - \lambda_0}{\lambda} & \text{if } \lambda_0 < \lambda < \lambda_f \\ 1 & \text{if } \lambda_f < \lambda \end{cases} \quad (25)$$

Following the derivation in [Rezaei et al. \(2017\)](#) and in analogy to the reduced dissipation inequality (cf. Section 2.1), the dissipation related to the cohesive zone is obtained as

$$D_{CZ} = Y_{CZ}\dot{D}_{CZ} = \frac{1}{2}k_0\lambda^2\dot{D}_{CZ} \geq 0. \quad (26)$$

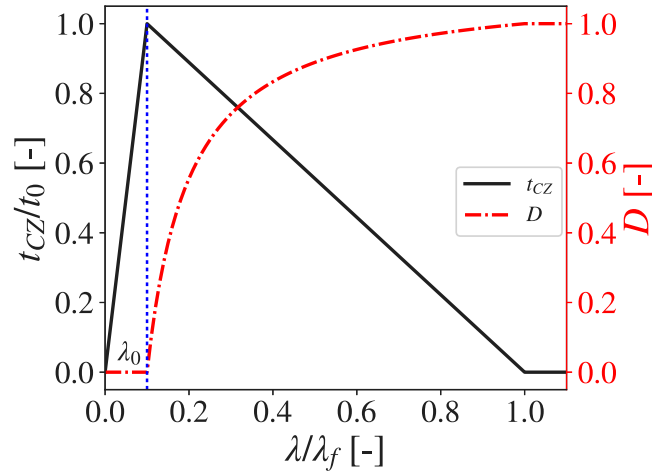


Fig. 2. Visualization of traction and damage evolution of a bilinear traction-separation law; traction and separation are normalized to maximum traction t_0 and failure separation λ_f .

Here, $Y_{CZ} = -\partial\psi_{CZ}/\partial D_{CZ}$ is the damage driving force of the interface. Since $\frac{1}{2}k_0\lambda^2 \geq 0$ is fulfilled at any time, the inequality can be reduced to

$$\dot{D}_{CZ} \geq 0, \quad (27)$$

showing that no damage healing takes place on the interface.

3. Finite element implementation

In the following, we focus on the implementations of the material models introduced in Section 2 as user-defined elements into the finite element software FEAP.

3.1. Implementation of bulk

First, the implementation of the balance of linear momentum and the additional micromorphic balance – as introduced in Section 2.1 – are presented. While only the key aspects are described here, more detailed investigations can be found, e.g. in Holthusen et al. (2020) and Fassin et al. (2019b) or Barfusz et al. (2021).

The domain Ω_0^h is introduced as an approximation of the referential body Ω_0 . The domain Ω_0^h in turn is split into sub domains Ω_0^e , which will later on be referred to as elements. Index e refers to a unique element identifier (e.g. element number):

$$\Omega_0 \approx \Omega_0^h = \bigcup_{e=1}^{n_{el}} \Omega_0^e \quad (28)$$

with n_{el} being the total number of elements. Furthermore, shape functions $\mathbf{N}(\mathbf{X})$ (with $\mathbf{N}(\mathbf{X})$ being a matrix with dimensions adjusted to the field interpolated) are introduced to approximate the field variables $\mathbf{u}(\mathbf{X})$ and $\mathbf{D}^X(\mathbf{X})$ from point-wise defined nodal values $\mathbf{u}^e$ and $\mathbf{D}^{X,e}$.

$$\mathbf{u}(\mathbf{X}) = \mathbf{N}(\mathbf{X}) \cdot \mathbf{u}^e; \quad \mathbf{D}^X(\mathbf{X}) = \mathbf{N}(\mathbf{X}) \cdot \mathbf{D}^{X,e} \quad (29)$$

The same type of trilinear shape functions are used for the displacement and micromorphic damage field, respectively.

As shown in Poggenpohl et al. (2020), the corresponding weak forms (g_u and g_D) of the balance equations are found by multiplying both parts of Eq. (14) with their corresponding test functions ($\delta\mathbf{u}$ and $\delta\mathbf{D}^X$) and integration over the volume. The weak forms are then linearized with respect to the corresponding increments ($\Delta\delta\mathbf{u}$ and $\Delta\delta\mathbf{D}^X$):

$$g_u := \int_{\Omega_0} \mathbf{S} : \delta\mathbf{E} \, dV - \int_{\Omega_0} \mathbf{f}_0 \cdot \delta\mathbf{u} \, dV - \int_{\partial_n\Omega_0} \mathbf{t}_0 \cdot \delta\mathbf{u} \, dA = 0 \quad (30)$$

$$g_D := \int_{\Omega_0} \delta\mathbf{D}^X \cdot \mathbf{H}^X [\mathbf{D} - \mathbf{D}^X] \, dV - \int_{\Omega_0} \nabla_0[\delta\mathbf{D}^X] \cdot [El^2 \nabla_0 \mathbf{D}^X] \, dV = 0 \quad (31)$$

$$\Delta g_u = \int_{\Omega_0} \Delta\mathbf{S} : \delta\mathbf{E} \, dV + \int_{\Omega_0} \mathbf{S} \cdot \Delta\delta\mathbf{E} \, dV \quad (32)$$

$$\Delta g_D = \int_{\Omega_0} \delta D^\chi H^\chi [\Delta D - \Delta D^\chi] dV - \int_{\Omega_0} \nabla_0 [\delta D^\chi] \cdot [El^2 \nabla_0 [\Delta D^\chi]] dV \quad (33)$$

Here, δE is defined as $\delta E = \frac{1}{2} [F^T \cdot \nabla_0(\delta \mathbf{u}) + \nabla_0(\delta \mathbf{u})^T \cdot \mathbf{F}]$, while ΔS and ΔD describe the sensitivity of stress and damage with respect to changes in $\mathbf{u}$ and D^χ , respectively. A more detailed description can be found in Poggenpohl et al. (2020).

The integrals of Eqs. (30) to (33) can be divided into the aforementioned element domains Ω_0^e . Hence, the weak forms and their corresponding linearizations read:

$$g_u^h = \bigcup_{e=1}^{n_e} \delta \mathbf{u}^{e,T} \cdot \mathbf{r}_u^e, \quad \Delta g_u^h = \bigcup_{e=1}^{n_e} \delta \mathbf{u}^{e,T} \cdot [\mathbf{K}_{u,u}^e \cdot \Delta \mathbf{u}^e + \mathbf{K}_{u,D^\chi}^e \cdot \Delta D^{\chi,e}] \quad (34)$$

$$g_D^h = \bigcup_{e=1}^{n_e} \delta D^{\chi,e} \cdot \mathbf{r}_D^e, \quad \Delta g_D^h = \bigcup_{e=1}^{n_e} \delta D^{\chi,e,T} \cdot [\mathbf{K}_{D^\chi,u}^e \cdot \Delta \mathbf{u}^e + \mathbf{K}_{D^\chi,D^\chi}^e \cdot \Delta D^{\chi,e}] \quad (35)$$

In above Eqs. (34) and (35) $\mathbf{r}_u^e$ and $\mathbf{r}_D^e$ represent the element residual vectors, while $\mathbf{K}_{u,u}^e$, $\mathbf{K}_{u,D^\chi}^e$, $\mathbf{K}_{D^\chi,u}^e$ and $\mathbf{K}_{D^\chi,D^\chi}^e$ refer to the element stiffness matrices. A detailed discussion of the residual vectors and stiffness matrices can be found in Poggenpohl et al. (2020).

Within a Newton–Raphson iteration, the sum of the weak form and its linearization have to be zero. Furthermore, it has to be taken into account that $\delta \mathbf{u}$ and δD^χ can take arbitrary (but nonzero) values. Thus, the element stiffness matrix and element residual vector can be found:

$$\begin{pmatrix} \mathbf{K}_{u,u}^e & \mathbf{K}_{u,D^\chi}^e \\ \mathbf{K}_{D^\chi,u}^e & \mathbf{K}_{D^\chi,D^\chi}^e \end{pmatrix} \begin{pmatrix} \Delta \mathbf{u}^e \\ \Delta D^{\chi,e} \end{pmatrix} = - \begin{pmatrix} \mathbf{r}_u^e \\ \mathbf{r}_D^e \end{pmatrix} \quad (36)$$

Within a finite element code, these element-wise equation systems are then assembled to form the global system of equations, which in turn is solved using the boundary conditions defined.

Finally, to ensure numerical stability and to prevent snapback behavior, an artificial viscosity as introduced in Fassin et al. (2019a) or Holthausen et al. (2020) is utilized. Here, the viscosity term is added to the stress and stiffness of the mechanical part on the Gauss point level, which are computed from:

$$\mathbf{S}_{vis} = \eta \frac{\Delta E}{\Delta t} \quad (37)$$

$$\mathbb{C}_{vis} = \frac{\eta}{\Delta t} \mathbb{I} \quad (38)$$

In the above, Δ denotes the increment of the current step with respect to the previous calculated step and $\mathbb{I}$ is the fourth order identity tensor. It shall be noted that the strain increment ΔE is calculated from

$$\Delta E = \mathbf{F}_{n+1}^T \cdot \mathbf{F}_{n+1} - \frac{1}{2} \mathbf{F}_n^T \cdot \mathbf{F}_{n+1} - \frac{1}{2} \mathbf{F}_{n+1}^T \cdot \mathbf{F}_n \quad (39)$$

to give consistent results. Here, $\mathbf{F}_{n+1}^T$ denotes the deformation gradient of the current time step while $\mathbf{F}_n^T$ denotes the deformation gradient of the previous time step.

3.2. Implementation of the interface

The implementation of the interface response into cohesive zone elements (CZE) is based on the formulation introduced in Rezaei et al. (2017). The CZE is introduced as an eight node element between two phases of a body (denoted here as $\Omega_{0,1}$ and $\Omega_{0,2}$) as indicated in Fig. 3. Here, the surface of one phase is indicated by Γ_+ , while the surface of the other phase is indicated by Γ_- .

For a better approximation of quantities, a middle plane is introduced:

$$\mathbf{x}_{mid} = \frac{1}{2}(\mathbf{x}_+ + \mathbf{x}_-) \quad (40)$$

Here, $\mathbf{x}_+$ and $\mathbf{x}_-$ are the position vectors (in reference configuration) on each side of the cohesive zone. For a zero thickness CZE the position vectors of the undeformed configuration for each side of the CZ are identical (i.e. $\mathbf{x}_+ \equiv \mathbf{x}_-$). However, due to numerical reasons, the CZs used in this publication initially have a – small – thickness. Gap and traction vectors are calculated with respect to this middle plane. Details on how to calculate the effective traction and gap vectors of the middle plane can be found in Rezaei et al. (2017).

The CZE is restricted to a certain normal direction of the local coordinate system. However, within the meshes used here, the orientation of the local coordinate system cannot be known a priori. Therefore, a renumbering of the nodes at the element level is introduced as suggested in Reese (2007) for solid shell elements.

Additionally, the CZ element residual vector and stiffness matrix have to be extended to fit the shape of the residual vector and stiffness matrix of the bulk, respectively. It is assumed that the interface does not influence or is influenced by the micromorphic damage field. Therefore, it is sufficient to use the local form of the damage formulation here. The residuum and stiffness associated with the micromorphic damage field are zero:

$$\begin{pmatrix} \mathbf{K}_{u,u}^e & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \Delta \mathbf{u}^e \\ \Delta D^{\chi,e} \end{pmatrix} = - \begin{pmatrix} \mathbf{r}_u^e \\ \mathbf{0} \end{pmatrix} \quad (41)$$

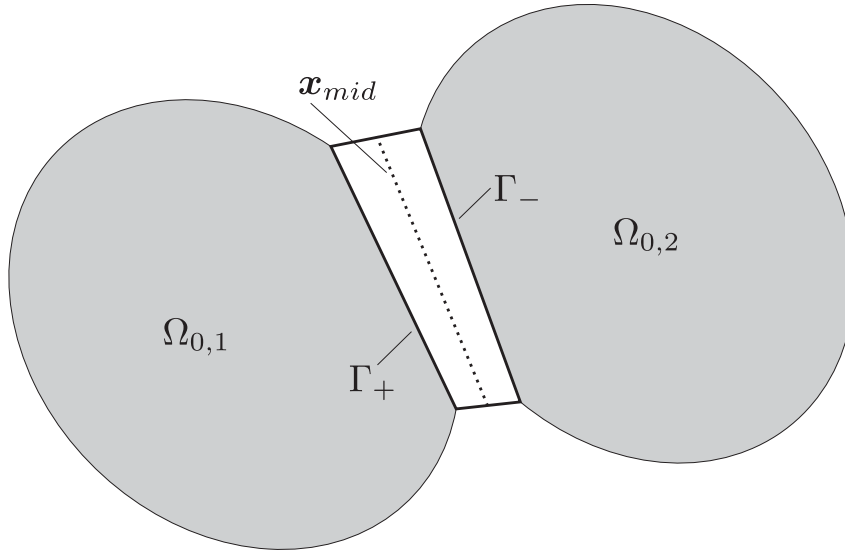


Fig. 3. Visualization of a cohesive zone element.

From Eq. (41) it can be seen that the CZE elements decouple the phases in case of the micromorphic variable. As will be shown later, the carbon fibers are assumed to have a much higher damage resistance than the surrounding matrix material and do not damage at the given loading. Hence, the effect of hypothesizing such a decoupled micromorphic damage field is assumed to be negligible.

4. Homogenization theory

In the following, the classical Hill homogenization approach will be enhanced for gradient-extended material models based on the micromorphic concept.

For that, Hill's approach is applied first on an RUC with perfect interfaces between fibers and matrix and without localization. Next, the influence of damaging interfaces – i.e. debonding between fibers and matrix – will be taken into account. Finally, the novel failure zone homogenization scheme tackling both, damage- and debonding-induced, softening is derived, leading to an advanced homogenization formulation.

In the following, a subscript M denotes quantities of the macro level, while m denotes quantities of the micro level.

4.1. Homogenization scheme accounting for gradient-extended (micromorphic) damage

Based on the hypothesized equivalence of virtual work density on the microscopic and macroscopic level, the Hill–Mandel condition Hill (1967, 1972) states that for a representative volume element the product of the averages is equal to the average of the products:

$$\delta \mathcal{W}_M = \{\delta \mathcal{W}_m\}_{\Omega_0} \iff \{\mathbf{P}_m\}_{\Omega_0} : \{\delta \mathbf{F}_m\}_{\Omega_0} = \{\mathbf{P}_m : \delta \mathbf{F}_m\}_{\Omega_0} \quad (42)$$

Here, $\mathbf{P}_m$ is the first Piola–Kirchhoff stress tensor of the micro scale, which is the work conjugate stress quantity to the deformation gradient of the micro scale $\mathbf{F}_m$. Further, $\{*\}_{\Omega_0}$ denotes the volume average, i.e.:

$$\{*\}_{\Omega_0} = \frac{1}{|\Omega_0|} \int_{\Omega_0} (*) \, dV, \quad (43)$$

where, $|\Omega_0|$ refers to the volume of Ω_0 . As described, e.g. in Geers et al. (2017), one can show that for certain sets of boundary conditions such as periodic boundary conditions, Eq. (42) implies that the stress of a macroscale material point, $\mathbf{P}_M$, is the same as the volume averaged stresses within the corresponding volume element on the micro-scale:

$$\mathbf{P}_M = \{\mathbf{P}_m\}_{\Omega_0} \quad (44)$$

Nevertheless, further investigations are necessary when considering gradient materials, as shown e.g. in Hirschberger et al. (2008), because for an accurate homogenization, the work conjugates of the gradient terms have to be taken into account as well. Hence, the contribution of micromorphic work conjugates a and b has to be considered, giving

$$\{\mathbf{P}_m\}_{\Omega_0} : \{\delta \mathbf{F}_m\}_{\Omega_0} = \{\mathbf{P}_m : \delta \mathbf{F}_m + a \delta D^\chi + b \cdot \nabla_0 \delta D^\chi\}_{\Omega_0}. \quad (45)$$

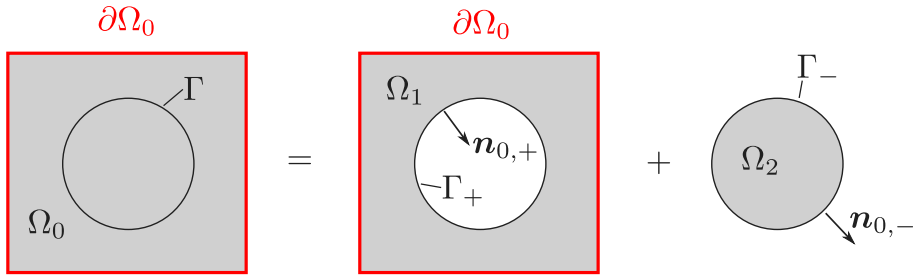


Fig. 4. Matrix (Ω_0) and embedded inclusion (Ω_2) separated in subdomains by a cohesive zone (Γ_+ and Γ_-).

However, according to Clasen et al. (2013), for micromorphic extended material models the virtual work done by the work conjugates for the entire body (Ω_0) vanishes. For completeness, the important steps are shown in the following.

Starting from the sum of all virtual work contributions in Eq. (45), the latter term of this equation can be rewritten by exploiting the product rule:

$$\begin{aligned} \text{Div}(\mathbf{b}\delta D^\chi) &= \text{Div}(\mathbf{b})\delta D^\chi + \mathbf{b} \cdot \nabla_0 D^\chi \\ \Leftrightarrow \mathbf{b} \cdot \nabla_0 \delta D^\chi &= \text{Div}(\mathbf{b}\delta D^\chi) - \text{Div}(\mathbf{b})\delta D^\chi \end{aligned} \quad (46)$$

Rearranging the micromorphic balance equation (Eq. (14)) yields:

$$\text{Div}(\mathbf{b}) = \mathbf{a} \quad (47)$$

Hence, application of Eqs. (10) and (13) to Eq. (15), the following condition for $\mathbf{b}$ at the boundary is achieved:

$$\mathbf{b} \cdot \mathbf{n}_0 = El^2 \nabla_0 D^\chi \cdot \mathbf{n}_0 = 0 \quad \text{on } \partial\Omega_0 \quad (48)$$

Then, the virtual work of the micromorphic extension can be rewritten using above Eqs. (46) to (48) and applying the divergence theorem:

$$\begin{aligned} \int_{\Omega_0} [a \delta D^\chi + \mathbf{b} \cdot \nabla_0 \delta D^\chi] dV &\stackrel{(46)}{=} \int_{\Omega_0} [a \delta D^\chi + \text{Div}(\mathbf{b}\delta D^\chi) - \text{Div}(\mathbf{b})\delta D^\chi] dV \\ &\stackrel{(47)}{=} \int_{\Omega_0} [a \delta D^\chi + \text{Div}(\mathbf{b}\delta D^\chi) - a \delta D^\chi] dV = \int_{\Omega_0} \text{Div}(\mathbf{b}\delta D^\chi) dV \\ &= \int_{\partial\Omega_0} \mathbf{b} \cdot \mathbf{n}_0 \delta D^\chi dA \stackrel{(48)}{=} 0 \end{aligned} \quad (49)$$

From Eq. (49) it becomes apparent that the classical Hill's theorem (42) applies even at the presence of micromorphic entities. Noticeably, this only holds true for homogenization over the whole computational domain and perfect interfaces.

4.2. Homogenization scheme accounting for debonding of non-perfect interfaces

In what follows, we derive the homogenization scheme including non-perfect interfaces based on the derivations provided in Geers et al. (2017). For RUCs of carbon fiber reinforced plastics, this refers to the interface of carbon fibers and epoxy matrix. Hence, the homogenization approach presented in the previous Section 4.1 has to be extended. For simplicity, one inclusion and its interface are considered here, but the methodology also holds for more inclusions. As can be seen in Fig. 4, the body Ω_0 is separated by the cohesive zone into the sub domains Ω_1 and Ω_2 with boundaries $\partial\Omega_0$, Γ_+ and Γ_- , respectively. Normal vectors $\mathbf{n}_{0,+}$ and $\mathbf{n}_{0,-}$ are introduced for each of the boundaries. Hence, the subscripts + and - denote quantities on the boundaries of Ω_1 and Ω_2 (i.e. on the different sides of the interface), respectively.

Based on the separation in Fig. 4, the volume integral of the stress can be divided into volume integrals of the two subdomains which, in turn, can be transformed into three surface integrals:

$$\begin{aligned} \frac{1}{|\Omega_0|} \int_{\Omega_0} \mathbf{P}_m dV &= \frac{1}{|\Omega_0|} \left(\int_{\Omega_1} \mathbf{P}_m dV + \int_{\Omega_2} \mathbf{P}_m dV \right) \\ &= \frac{1}{|\Omega_0|} \left(\int_{\partial\Omega_0} \underbrace{\mathbf{n}_{0,0} \cdot \mathbf{P}_{m,0}^T}_{t_0} \otimes \mathbf{X}_{m,0} dA + \int_{\Gamma_+} \underbrace{\mathbf{n}_{0,+} \cdot \mathbf{P}_{m,+}^T}_{t_{0,+}} \otimes \mathbf{X}_{m,+} dA \right. \\ &\quad \left. + \int_{\Gamma_-} \underbrace{\mathbf{n}_{0,-} \cdot \mathbf{P}_{m,-}^T}_{t_{0,-}} \otimes \mathbf{X}_{m,-} dA \right) \end{aligned} \quad (50)$$

Within the surface integrals, the surface tractions $\mathbf{t} = 0$, $\mathbf{t}_{0,+}$ and $\mathbf{t}_{0,-}$ on the boundaries are introduced. Furthermore, since all quantities are defined in the reference configuration, it can be assumed that the boundaries Γ_+ and Γ_- coincide (i.e. $\Gamma_+ = \Gamma_- = \Gamma$). In addition, the stress continuity condition can be applied

$$\mathbf{P}_{m,+} = \mathbf{P}_{m,-} \quad (51)$$

and the normal vectors shall be consistent through all sub geometries

$$\mathbf{n}_0 = \mathbf{n}_{0,+} = -\mathbf{n}_{0,-} \quad (52)$$

Thus, the surface tractions on each side of the interface are equal in amount but opposite in direction. Consequently, Eq. (50) can be simplified to form

$$\begin{aligned} \frac{1}{|\Omega_0|} \int_{\Omega_0} \mathbf{P}_m \, dV &= \frac{1}{|\Omega_0|} \left(\int_{\partial\Omega_0} \mathbf{n}_0 \cdot \mathbf{P}_m^T \otimes \mathbf{X}_m \, dA + \int_{\Gamma} \mathbf{t} \otimes \mathbf{X}_m \, dA + \int_{\Gamma} -\mathbf{t} \otimes \mathbf{X}_m \, dA \right) \\ &= \frac{1}{|\Omega_0|} \left(\int_{\partial\Omega_0} \mathbf{n}_0 \cdot \mathbf{P}_m^T \otimes \mathbf{X}_m \, dA \right) \end{aligned} \quad (53)$$

As a result, the introduction of an interface does not affect the volume average of the stresses.

In analogy to $\mathbf{P}$, the following is found for $\mathbf{F}$:

$$\begin{aligned} \frac{1}{|\Omega_0|} \int_{\Omega_0} \mathbf{F}_m \, dV &= \frac{1}{|\Omega_0|} \left(\int_{\Omega_1} \mathbf{F}_m \, dV + \int_{\Omega_2} \mathbf{F}_m \, dV \right) \\ &= \frac{1}{|\Omega_0|} \left(\int_{\partial\Omega_0} \mathbf{x} \otimes \mathbf{n}_0 \, dA + \int_{\Gamma_+} \mathbf{x} \otimes -\mathbf{n}_0 \, dA + \int_{\Gamma_-} \mathbf{x} \otimes \mathbf{n}_0 \, dA \right) \end{aligned} \quad (54)$$

In contrast to the traction, a continuity condition does not exist for the deformation field. Therefore, the displacement jump over the interface

$$\mathbf{g} = \mathbf{x}_- - \mathbf{x}_+ \quad (55)$$

is defined. Consequently, the displacement jump over the interface is equal to the gap vector of the CZE as described in Section 2.2. Hence, Eq. (54) can be rewritten, yielding:

$$\underbrace{\frac{1}{|\Omega_0|} \int_{\Omega_0} \mathbf{F}_m \, dV}_{\mathbf{F}_{bulk}} = \underbrace{\frac{1}{|\Omega_0|} \int_{\partial\Omega_0} \mathbf{x} \otimes \mathbf{n}_0 \, dA}_{\mathbf{F}_{ext}} - \underbrace{\frac{1}{|\Omega_0|} \int_{\Gamma} \mathbf{g} \otimes \mathbf{n}_0 \, dA}_{\mathbf{F}_{CZ}} \quad (56)$$

It shall be noted that coinciding Γ_+ and Γ_- is used here. Then, the displacement gradient averaged over the bulk ($\mathbf{F}_{bulk}$) is the same as the displacement gradient measured over the external boundary ($\mathbf{F}_{ext}$) subtracted by the displacement gradient of the interface ($\mathbf{F}_{CZ}$). Rearranging Eq. (56) yields

$$\mathbf{F}_{ext} = \mathbf{F}_{bulk} + \mathbf{F}_{CZ} \quad (57)$$

which states that the (externally measured) displacement is composed of the displacement of the bulk as well as the displacement jump of the interface.

From these derivations for stress and deformation gradient, a homogenization approach at the presence of interfaces can be derived accordingly for the virtual work of the bulk:

$$\begin{aligned} \frac{1}{|\Omega_0|} \int_{\Omega_0} \mathbf{P}_m : \delta \mathbf{F}_m \, dV &= \frac{1}{|\Omega_0|} \left(\int_{\Omega_1} \mathbf{P}_m : \delta \mathbf{F} \, dV + \int_{\Omega_2} \mathbf{P}_m : \delta \mathbf{F} \, dV \right) \\ &= \frac{1}{|\Omega_0|} \left(\int_{\partial\Omega_0} \mathbf{n}_0 \cdot \mathbf{P}_m^T \delta \mathbf{x} \, dA + \int_{\Gamma_+} -\mathbf{n}_0 \cdot \mathbf{P}_m^T \delta \mathbf{x} \, dA + \int_{\Gamma_-} \mathbf{n}_0 \cdot \mathbf{P}_m^T \delta \mathbf{x} \, dA \right) \\ &= \frac{1}{|\Omega_0|} \int_{\partial\Omega_0} \mathbf{n}_0 \cdot \mathbf{P}_m^T \delta \mathbf{x} \, dA - \frac{1}{|\Omega_0|} \int_{\Gamma} \overbrace{\mathbf{n}_0 \cdot \mathbf{P}_m^T}^{\mathbf{t}} \delta \mathbf{g} \, dA \end{aligned} \quad (58)$$

In case of the virtual work, the analogy to the deformation gradient becomes apparent. Above equation can be rearranged, giving:

$$\underbrace{\frac{1}{|\Omega_0|} \int_{\partial\Omega_0} \mathbf{n}_0 \cdot \mathbf{P}_m^T \cdot \delta \mathbf{x} \, dA}_{\delta \mathcal{W}_{ext}} = \underbrace{\frac{1}{|\Omega_0|} \int_{\Omega_0} \mathbf{P}_m : \delta \mathbf{F}_m \, dV}_{\delta \mathcal{W}_{bulk}} + \underbrace{\frac{1}{|\Omega_0|} \int_{\Gamma} \mathbf{t} \cdot \delta \mathbf{g} \, dA}_{\delta \mathcal{W}_{CZ}} \quad (59)$$

Here, the external virtual work, which is equal to the macroscopic virtual work, is composed of the virtual work of the bulk and the virtual work of the interface. Therefore, a homogenization scheme also has to account for the contributions of the interface.

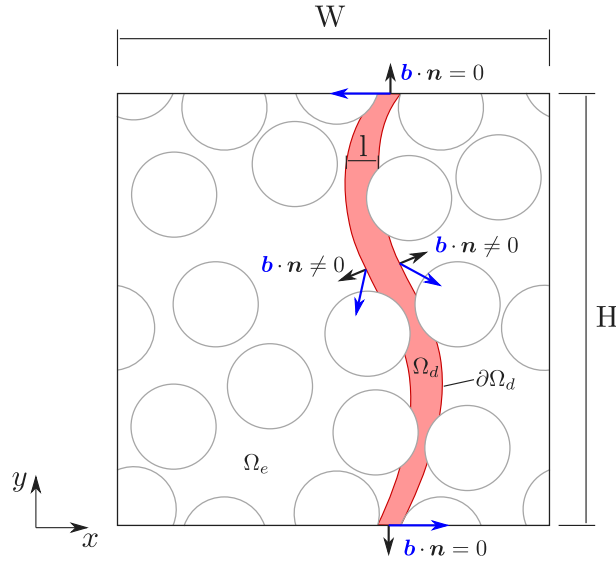


Fig. 5. Exemplary geometry with failure zone Ω_d (red) and elastically unloading zone Ω_e ; normal vector ($\mathbf{n}$) and micromorphic stress vector ($\mathbf{b}$) are shown at the boundary of Ω_d . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.3. Failure zone homogenization scheme with gradient-extended (micromorphic) damage and debonding

In what follows, the results from previous Sections 4.1 and 4.2 are combined. Moreover, the failure zone averaging method introduced by Nguyen et al. (2010), which ensures size-independent results even in the softening regime, is extended accordingly.

It was shown e.g. in Gitman et al. (2007) that classical Hill–Mandel homogenization is not applicable at the presence of localization phenomena. This is related to a mismatch in the dimensions between the elastic volume (scaling with $W \times H$ per unit depth) and the dissipating volume (scaling only with $W \times \ell$ or $H \times \ell$ per unit depth) as visualized in Fig. 5. This mismatch causes the RUC to show a more brittle behavior with increasing size as will be shown in Section 6.3.

The failure zone averaging as introduced in Nguyen et al. (2010) has proven to overcome the aforementioned dimension mismatch by applying the volume averaging only on the actively damaging domain Ω_d which is defined as

$$\Omega_d := \{\mathbf{x} \in \Omega_0 \mid \dot{D}(\mathbf{x}) > 0\}. \quad (60)$$

A representation of the failure zone is shown in Fig. 5, where Ω_e describes the elastically unloading zone. Analogously, a domain of actively damaging interfaces can be identified as well:

$$\Gamma_d := \{\mathbf{x} \in \Gamma_0 \mid \dot{D}(\mathbf{x}) > 0\} \quad (61)$$

It was shown in Nguyen et al. (2010) and following publications (e.g. Nguyen et al., 2011 or Turteltaub and de Jong, 2019) that this averaging technique resolves the issue of dimension mismatch. Thus, a representative size for the RUCs can be found.

As derived in Eq. (49), the virtual work of the micromorphic constituents vanishes when averaging over the whole domain. But as can be seen in Eq. (48), this depends on the boundary condition applied on $\nabla_0 D^\chi$. In general, as indicated in Fig. 5, the boundaries $\partial\Omega_d$ and $\partial\Omega_0$ do not coincide, and therefore, the micromorphic quantities do not vanish. Consequently, the Hill condition in this case must be adjusted:

$$\begin{aligned} \{\mathbf{P}_m\}_{\Omega_d} : \{\delta \mathbf{F}_m\}_{\Omega_d} &= \frac{1}{|\Omega_d|} \int_{\Omega_d} [\mathbf{P}_m : \delta \mathbf{F}_m + a \delta D^\chi + \mathbf{b} \cdot \nabla_0 \delta D^\chi] \, dV \\ &= \frac{1}{|\Omega_d|} \int_{\Omega_d} [\mathbf{P}_m : \delta \mathbf{F}_m + \text{Div}(\mathbf{b} \delta D^\chi)] \, dV \\ &= \frac{1}{|\Omega_d|} \int_{\partial\Omega_d} [\mathbf{n}_0 \cdot \mathbf{P}_m^T \cdot \delta \mathbf{x}_m + \mathbf{n}_0 \cdot \mathbf{b} \delta D^\chi] \, dA \end{aligned} \quad (62)$$

In consequence, the mechanical virtual work generally is accompanied by the micromorphic virtual work. In line with Geers et al. (2017), the following relation for $\delta \mathbf{x}_m$ is introduced

$$\delta \mathbf{x}_m = \{\delta \mathbf{F}_m\}_{\Omega_d} \cdot \mathbf{X}_m + \delta \mathbf{w}_m \quad (63)$$

assuming that the virtual deformation of a material point, $\delta \mathbf{x}_m$, can be described via a virtual average deformation, $\{\delta \mathbf{F}_m\}_{\Omega_d} \cdot \mathbf{X}_m$, and a virtual fluctuation term, $\delta \mathbf{w}_m$. Application of Eq. (63) on Eq. (62) yields:

$$\{\mathbf{P}_m\}_{\Omega_d} : \{\delta \mathbf{F}_m\}_{\Omega_d} = \frac{1}{|\Omega_d|} \int_{\partial\Omega_d} [\mathbf{n}_0 \cdot \mathbf{P}_m^T \cdot (\{\delta \mathbf{F}_m\}_{\Omega_d} \cdot \mathbf{X}_m + \delta \mathbf{w}_m) + \mathbf{n}_0 \cdot \mathbf{b} \delta D^\chi] \, dA$$

$$\begin{aligned}
&= \frac{1}{|\Omega_d|} \left(\underbrace{\int_{\Omega_d} \mathbf{P}_m dV}_{=0?} + \underbrace{\int_{\partial\Omega_d} \mathbf{n}_0 \cdot \mathbf{P}_m^T \otimes \mathbf{X}_m dA : \{\delta \mathbf{F}_m\}_{\Omega_d}}_{=0?} + \underbrace{\int_{\partial\Omega_d} \mathbf{n}_0 \cdot \mathbf{P}_m^T \cdot \delta \mathbf{w}_m dA}_{=0?} \right. \\
&\quad \left. + \underbrace{\int_{\partial\Omega_d} \mathbf{n}_0 \cdot \mathbf{b} \delta D^\times dA}_{\neq 0} \right) \quad (64)
\end{aligned}$$

From Eq. (64) it becomes apparent that the virtual work of the bulk in the actively damaging zone, Ω_d , can be decomposed into three parts which are defined on the boundary of Ω_d . As given in Geers et al. (2017), the first term leads to the homogenization relation of the stress, Eq. (44), but only if the second term vanishes. It is shown in the latter publication that the second term vanishes for certain boundary conditions (e.g. periodic or fixed boundary conditions). However, as indicated in Eq. (64), it is questionable whether these boundary conditions also apply on the boundary of Ω_d . Likewise, the third term does not vanish in the case of failure zone averaging. Therefore, a homogenization condition for the Stress would have to account for this energetic contribution additionally to the contribution of the mechanical parts.

As shown in Eq. (64), it has to be determined whether the second term vanishes. Therefore, the surface integral is transformed into a volume integral giving

$$\int_{\Omega_d} (\text{Div}(\mathbf{P}_m) \cdot \delta \mathbf{w}_m + \mathbf{P}_m : \nabla_0(\delta \mathbf{w}_m)) dV = 0. \quad (65)$$

Here, the first term vanishes since $\text{Div}(\mathbf{P}_m) = 0$ holds for static problems without body forces. Then, the virtual fluctuations $\delta \mathbf{w}_m$ can be rewritten using reformulated Eq. (63):

$$\int_{\Omega_d} \mathbf{P}_m : \nabla_0(\delta \mathbf{x}_m - \{\delta \mathbf{F}_m\}_{\Omega_d} \cdot \mathbf{X}_m) dV = 0 \quad (66)$$

Rearranging above equation yields

$$\begin{aligned}
\int_{\Omega_d} \mathbf{P}_m : \nabla_0(\delta \mathbf{x}_m) dV &= \int_{\Omega_d} \mathbf{P}_m : \nabla_0(\{\delta \mathbf{F}_m\}_{\Omega_d} \cdot \mathbf{X}_m) dV \\
&\Leftrightarrow \int_{\Omega_d} \mathbf{P}_m : \delta \mathbf{F}_m dV = \int_{\Omega_d} \mathbf{P}_m dV : \{\delta \mathbf{F}_m\}_{\Omega_d}, \quad (67)
\end{aligned}$$

where it was exploited that $\{\delta \mathbf{F}_m\}_{\Omega_d}$ is constant in Ω_d . Eq. (67) divided by Ω_d gives

$$\{\mathbf{P}_m\}_{\Omega_d} : \{\delta \mathbf{F}_m\}_{\Omega_d} = \{\mathbf{P}_m : \delta \mathbf{F}_m\}_{\Omega_d}, \quad (68)$$

which is equal to the equation investigated in Nguyen et al. (2011). There, Eq. (68) was substantiated with numerical results. As shown here, this equation is equal to the assumption that the second term in Eq. (64) vanishes. In Section 6.3 it is numerically investigated whether Eq. (68) holds true.

In Section 4.2 it was shown that the virtual work of the bulk must be extended by the virtual work of the interface. Similarly, this also holds true for the failure zone averaging:

$$\begin{aligned}
\{\mathbf{P}_m\}_{\Omega_d} : \{\delta \mathbf{F}_m\}_{\Omega_d} &= \frac{1}{|\Omega_d|} \left(\int_{\partial\Omega_d} \mathbf{n}_0 \cdot \mathbf{P}_m^T \otimes \mathbf{X}_m dA : \delta \mathbf{F}_M + \int_{\partial\Omega_d} \mathbf{n}_0 \cdot \mathbf{P}_m^T \cdot \delta \mathbf{w}_m dA \right. \\
&\quad \left. + \int_{\partial\Omega_d} \mathbf{n}_0 \cdot \mathbf{b} \delta D^\times dA + \int_{\Gamma_d} \mathbf{t} \cdot \delta \mathbf{g} dA \right) \quad (69)
\end{aligned}$$

Likewise to the extension in Eq. (59), the virtual work in Eq. (64) has been extended by the energetic contribution of the actively damaging interfaces.

5. Generating repeating unit cells (RUCs)

The material of interest was long fiber reinforced plastic. It was assumed that the fibers were perfect cylinders and no inhomogeneities appeared along the fiber direction. The diameter of the fibers was assumed constant at $7 \mu\text{m}$ which is in accordance with e.g. Makeev et al. (2019). The geometries of the repeating unit cells were generated using three different approaches:

1. A regular pattern with manually defined fiber distribution,
2. the so-called *randomized* method, and
3. the *random sequential absorption* (RSA) method (see, e.g. Illian et al., 2008 or Schneider et al., 2016).

The latter two methods were used to automatically generate random fiber distributions. The volume fraction of the fibers was kept the same for all RUCs and was approximately 53.5%. A mesh with one layer (with a constant thickness of $1 \mu\text{m}$) of three dimensional, eight node, hexahedral elements was used for both, bulk and interface. Furthermore, height and width of the RUCs were chosen to be equal.

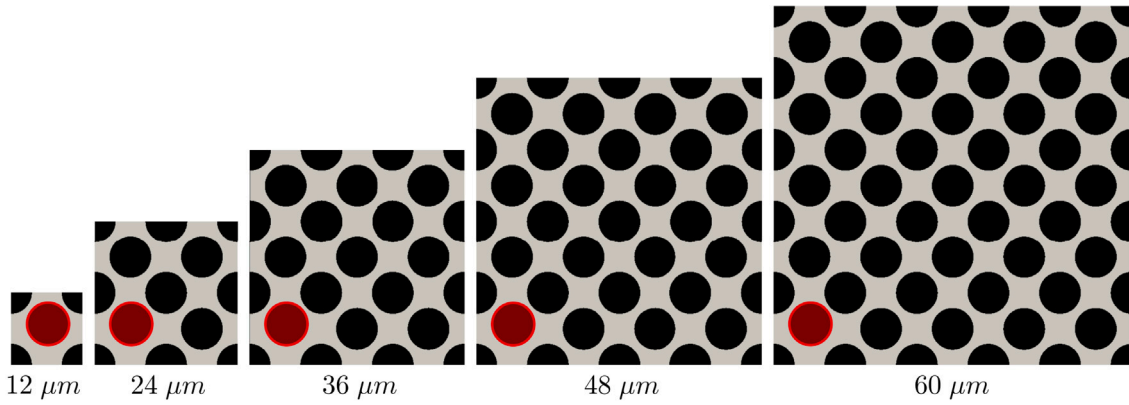


Fig. 6. Different sizes of RUCs used for a regular grid with imperfection (marked in red); RUC sizes range from $12\ \mu\text{m} \times 12\ \mu\text{m} \times 1\ \mu\text{m}$ (left) to $60\ \mu\text{m} \times 60\ \mu\text{m} \times 1\ \mu\text{m}$ (right). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

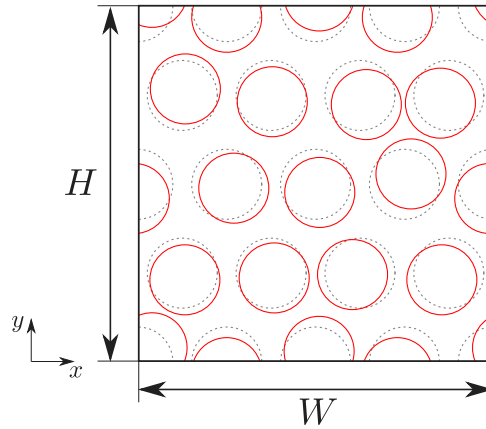


Fig. 7. Visualization of the generation of a *randomized* RUC geometry; solid black: RUC boundary, dashed: original regular distribution of fibers, red: final distribution of fibers. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The RUCs resulting from the *regular* grid resemble the hexagonal close packing as visualized in Fig. 6. Noticeably, the fibers at the lower left corner (marked in red) were shifted with respect to their position in a regular grid. This imperfection was applied to trigger damage evolution and avoid numerical instabilities due to bifurcation.

The *randomized* method for geometry generation was based on the regular grid, where a random shift was applied to each of the fibers as visualized in Fig. 7. In addition, the shift applied to each of the fibers was restricted to prevent overlapping fibers.

Another set of RUCs was generated using the *random sequential absorption (RSA)* method. As described in Illian et al. (2008), Schneider et al. (2016), fibers were sequentially placed into the RUC with random coordinates for each fiber, where only those fibers were accepted that did not overlap any of the already existing ones (see 8). To prevent narrow gaps, new fibers that were too close to already existing fibers were rejected in this approach as well. New fibers were generated until the desired amount of fiber volume fraction was reached. RUCs generated by this approach were referred as *RSA* RUCs.

Comparison of Figs. 7 and 8 showed that for the *RSA* generation method a higher degree of clustering occurred. *Randomized* RUCs in general showed a more even fiber distribution compared to *RSA* RUCs. Furthermore, fibers which overlapped the edges of the RUC were mirrored to the other side to create a repetitive geometry.

In Fig. 9, the deformed mesh of a *RSA* RUC with periodic boundary conditions at tensile loading is shown. The local fluctuations of the current configuration, which are mirrored at the edges (as marked here for the left and right edge) become visible. The periodic boundary conditions can be described via

$$\mathbf{x}_r^i - \mathbf{x}_l^i = \mathbf{F}_M \cdot (\mathbf{X}_r^i - \mathbf{X}_l^i), \quad (70)$$

where superscript i indicates a pair of nodes with the same reference coordinates in all but one direction. Exemplary, a pair of nodes for the left and right edges would have the same y - and z -coordinates but a constant difference in the x -coordinate. The macroscopic or far field deformation is indicated here by $\mathbf{F}_M$. It shall be noted that due to these boundary conditions, the displacement difference of each node pair is defined leaving enough freedom for the nodal deformation for local (repeating) fluctuations as indicated in

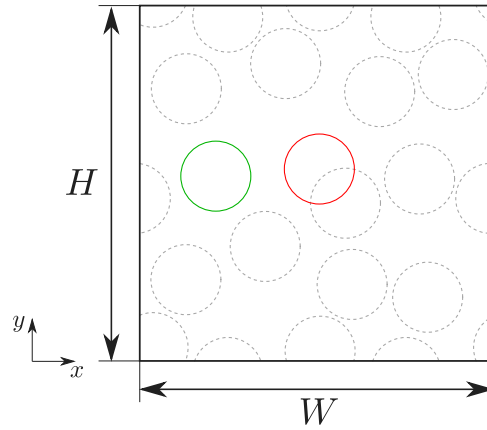


Fig. 8. Visualization of the generation of an RSA RUC geometry; solid black: RUC boundary, dashed: existing fibers, red: rejected fiber due to overlapping, green: accepted fiber. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

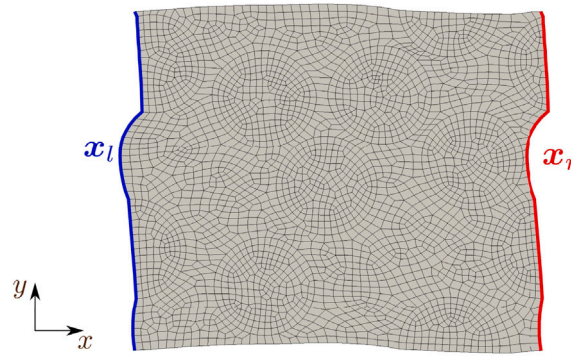


Fig. 9. Exemplary mesh and deformation of an $24\mu\text{m} \times 24\mu\text{m} \times 1\mu\text{m}$ RUC. Deformations magnified by a factor of 50.

Fig. 9. The periodic boundary conditions described here for the x -direction are applied in y - and z -direction as well. It shall be noted that no geometrical inhomogeneities are present in z -direction and all RUCs can be considered flat (i.e. the dimensions in x - and y -direction are much larger than in z -direction). Thus, the periodicity conditions in z -direction generally lead to material behavior comparable to plane strain. Within the software FEAP, the repeating boundary conditions were applied via the solution command *ELINK*, while the far field deformation gradient is applied via the solution command *PERIODIC, CAUCHY*. For the latter, the macroscopic displacement gradient ($\mathbf{H}_M = \mathbf{F}_M - \mathbf{I}$) is defined directly, while within FEAP displacement boundary conditions were derived from this input.

6. Numerical results

All RUCs were loaded with far field stretches in tensile direction (i.e. Mode I), while periodic boundary conditions were applied. It shall be emphasized that the homogenization procedure is applicable in the case of three dimensional RUCs as well. Due to the high numerical costs and challenges in mesh generation, the present study is restricted to the two dimensional case using one eight node element over the thickness of the RUC. Unless specifically stated, [Table 1](#) lists the material parameters used. Parameters were taken from [Poggenpohl et al. \(2020\)](#), where they had been fitted to experimental data.

Parameter l_1 was changed to fit the new length scales of the RUCs simulated. Here, parameter l_1 is treated as a numerical parameter and part of the micromorphic extension. In principle, this parameter could become physically relevant, if information about the localization zone width at the microscale is provided (e.g. via micro CT- or SEM scans). Generally, to high values for l_1 would lead to an even distribution of damage and thus, no localization band would form. In contrast, if parameter l_1 is chosen to small, the failure zone shrinks and a higher mesh density is needed in order to achieve mesh convergence. Here, parameter l_1 is chosen in a way, that a reasonable localization zone develops.

Here, parameter l is used as a numerical parameter to create a satisfactory failure zone width. If the parameter is chosen to large, all material points will have the same damage variable and no failure zone will develop. If parameter l is chosen to small, the failure zone shrinks and a higher mesh density is needed in order to achieve mesh convergence. In principle, in situ measurements

Table 1

Material parameters for the resin matrix (index 1) and carbon fibers (index 2). If no index is used, the same parameter was applied to both constituents.

	Parameter	Value	Parameter	Value	Parameter	Value
Matrix	λ_1	5308 MPa	K_1	6.5 MPa		
	μ_1	3538 MPa	H_1^x	2×10^6 MPa		
	$Y_{0,1}$	0.0672 MPa	l_1	6.4×10^{-4} mm		
Fiber	λ_2	423 077 MPa	K_2	0 MPa		
	μ_2	84 615 MPa	H_2^x	1 MPa		
	$Y_{0,2}$	1×10^6 MPa	l_2	0.0 mm		
Matrix & Fiber	h_{ic}	1	η	1×10^{-3} MPa s	n	2

Table 2

Work densities, maximum stresses, and relative errors according to (72) for different mesh sizes.

Mesh size [μm]	$\mathcal{W}_{tot}$ [J mm ⁻³]	$e_{rel,\mathcal{W}}$ [%]	σ_{max} [MPa]	$e_{rel,\sigma}$ [%]
2.00	1.516	9.73	137.49	1.82
1.00	1.425	3.16	135.76	0.54
0.50	1.383	0.10	135.14	0.08
0.25	1.381	–	135.02	–

of the localization zone of real specimen at the micro scale, e.g. micro CT or SEM scans, could give physical meaning to the length scale parameter l .

The fibers were modeled using an elastic, isotropic, material model. This simplification was made since the fibers were stiff compared to the matrix material and did not damage which is in accordance to, e.g. Fantoni et al. (2020), Guillén-Hernández et al. (2020) or Liu et al. (2021). Also, since the investigated effects were driven by and restricted to the behavior of the compliant matrix material the transverse isotropy of the fibers was neglected. To achieve the elastic response, the damage threshold of the fibers was chosen very high. Additionally, the micromorphic parameters of the fibers were chosen in a way that they do not influence the micromorphic field of the matrix.

In the following sections, the theoretical considerations will be underpinned by numerical examples. This includes:

- In the next section, a mesh convergence study is performed showing the effective treatment of mesh size sensitivity.
- Afterwards, the influence of the artificial viscosity on numerical results is presented.
- Thereafter, results obtained using the standard Hill's approach and the failure zone averaging method are compared.
- Next, a statistical investigation of the effect of generation algorithms on RUC response is given. Here, RUCs generated by the randomized and RSA approach are compared.
- Further, five different RUC sizes are compared in a statistical analysis of the deviations between different realizations of each RUC size.
- Then, micromorphic power and micromorphic work of the same sets of RUCs are investigated separately as numerical evidence for the derivation in Section 4.3.
- Finally, a statistical investigation of ten RSA RUCs with damaging interfaces is shown. Here, the statistical evolution of the work of bulk and interface is investigated.

6.1. Mesh convergence study

For a mesh convergence study, an RUC of size $48 \mu\text{m} \times 48 \mu\text{m} \times 1 \mu\text{m}$ using a regular grid for the fiber distribution was generated. Here, different mesh sizes ranging from $0.25 \mu\text{m}$ to $2 \mu\text{m}$ were used. It shall be noted that only the characteristic element length in plane of the RUC was varied, while one element over the thickness was used in all RUC simulations. Stresses and stretches were computed as volume averages of the whole computational domain Ω_0 .

Fig. 10 shows stress–stretch curves for the RUC at different mesh densities. Mesh sizes of $0.5 \mu\text{m}$ and $0.25 \mu\text{m}$ show almost the same stress–stretch curves. As shown, all simulations were terminated at a far field deformation of $F_{xx} = 1.025$.

Additionally, for an objective interpretation of the results, the total work density ($\mathcal{W}_{tot}$) was calculated

$$\mathcal{W}_{tot} := \int \{\mathbf{P} : \dot{\mathbf{F}}\}_{\Omega_0} dt, \quad (71)$$

where for numerical time integration the Midpoint Rule was applied.

As a result, numerical values for the work density and stress maximum can be found in Table 2. Here, the finest mesh was assumed to show the converged solution, and the relative errors were computed with respect to these results, such that

$$e_{rel,\mathcal{W}} = \frac{\mathcal{W}_{tot} - \mathcal{W}_{tot}^{0.25 \mu\text{m}}}{\mathcal{W}_{tot}^{0.25 \mu\text{m}}} \quad e_{rel,\sigma} = \frac{\sigma_{max} - \sigma_{max}^{0.25 \mu\text{m}}}{\sigma_{max}^{0.25 \mu\text{m}}} \quad (72)$$

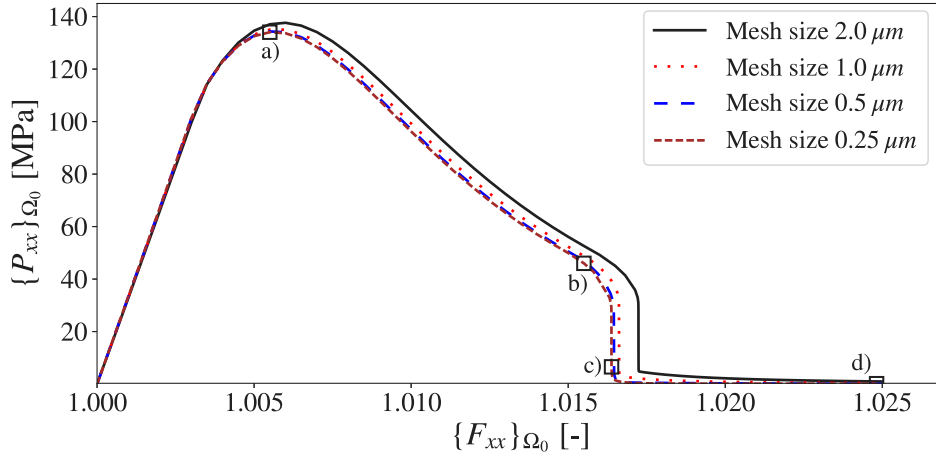


Fig. 10. Stress–stretch curves for different meshes of the regular RUC of size $48 \mu\text{m} \times 48 \mu\text{m} \times 1 \mu\text{m}$; damage contour plots of the fines mesh at the different stages a) to d) are shown in Figs. 11(a) to 11(d).

From Fig. 10 and Table 2 a clear convergence trend can be observed, highlighting that mesh dependence was successfully cured by the micromorphic extension of the damage formulation. Further, the numerical values in Table 2 suggest that a mesh size of $0.5 \mu\text{m}$ gave almost the same numerical value as the finest mesh in terms of work density and maximum stress. Therefore, a mesh size of $0.5 \mu\text{m}$ was used in the subsequent simulations.

In Figs. 11(a) to 11(d) damage contour plots also showing the deformed mesh are given. The different stages of deformation are indicated in Fig. 10. From Figs. 11(a) and 11(b) it becomes apparent that until load drop, damage was distributed homogeneously within the matrix. Fig. 11(c) shows the damage contour after load drop, where the localization zone has formed while Fig. 11(d) shows the deformed mesh at the end of simulation. From comparison of Figs. 11(c) and 11(d) it can be deduced that the damage contour does not change, but almost all deformation is accumulated in the heavily deformed localization zone. This result is underpinned by Figs. 11(e) and 11(f) where the actively damaging zone is plotted. While for $\{F_{xx}\}_{\Omega_0} = 1.0164$ a broad band is damaging, Ω_d is reduced to almost one layer of elements for $\{F_{xx}\}_{\Omega_0} = 1.025$.

In Fig. 12, the failure zone (i.e. the volume where $D \geq 0.8$ holds) including mesh deformation is shown. While from Fig. 11(d) a diffusive damage distribution might be deduced, the narrow failure (or localization) zone is seen in Fig. 12. As mentioned above, the width of the failure zone is controlled via parameter l_1 . Lower values of l_1 would result in less diffusion damage and a smaller failure zone.

The reduction of the actively damaging region to one layer of elements (Fig. 11(f)) might contradict the findings in Fig. 10 and Table 2. The reason for this inconsistency is found in the damage function of quadratic type as well as the fact, that almost no damage evolution takes place between $\{F_{xx}\}_{\Omega_0} = 1.0164$ and $\{F_{xx}\}_{\Omega_0} = 1.025$. Due to the quadratic failure function, a material point at $D = 0.995$ ($(1 - 0.995)^2 = 2.5 \times 10^{-5}$) has a much lower stiffness than a material point at $D = 0.98$ ($(1 - 0.98)^2 = 4 \times 10^{-4}$). A further increase in damage decreases the overall stress significantly. At the same time, the gradient between $D = 0.995$ and $D = 0.98$ is quite low leading to a decreased effect of the micromorphic extension. Thus, a damage zone as shown in Fig. 11(f) might develop way after full failure occurred. However, the mesh convergence shown in Fig. 10 and the actively damaging region right at the end of final failure (Fig. 11(e)) show that the solution is no longer mesh size dependent.

6.2. Influence of the artificial viscosity

In order to investigate the influence of artificial viscosity, the same RUC as for the mesh convergence study (Section 6.1) was used. Here, seven different simulations with viscosities ranging from 1×10^{-3} MPa s to 1×10^{-1} MPa s were calculated. Simulations with lower viscosities were carried out as well, but failed due to numerical instabilities. Again, stresses and stretches were computed as volume averages of the whole computational domain Ω_0 and simulations were terminated at a far field deformation of $F_{xx} = 1.025$. In Figs. 13(a) and 13(b), the stress–stretch curves and the evolution of the volume $|\Omega_d|$ of the actively damaging domain Ω_d normalized w.r.t. the overall volume $|\Omega_0|$ are shown, respectively.

As desired, the stresses–stretch curves only show a minor dependence on the artificial viscosity, such that the curves for $\eta = 1 \times 10^{-2}$ MPa s and $\eta = 1 \times 10^{-3}$ MPa s are almost identical. However, the curves of the actively damaging volume as presented in Fig. 13(b) show a quite significant dependence on the pseudo viscosity. From these results it was concluded that a viscosity of $\eta = 1 \times 10^{-3}$ MPa s leads to the best compromise between numerical stability and accuracy of the solution.

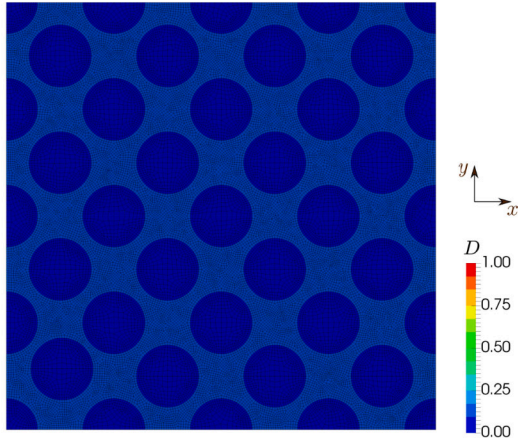
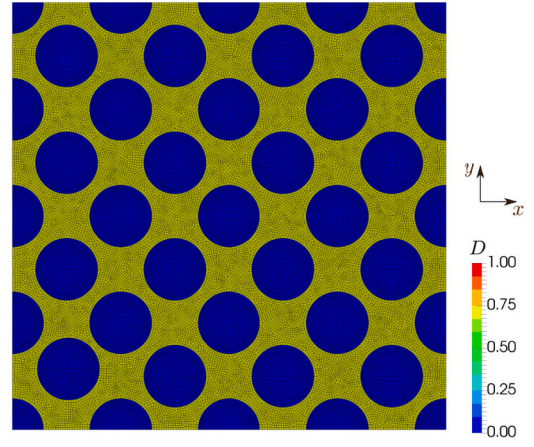
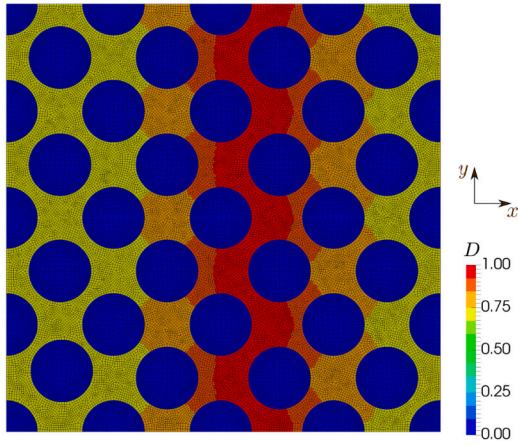
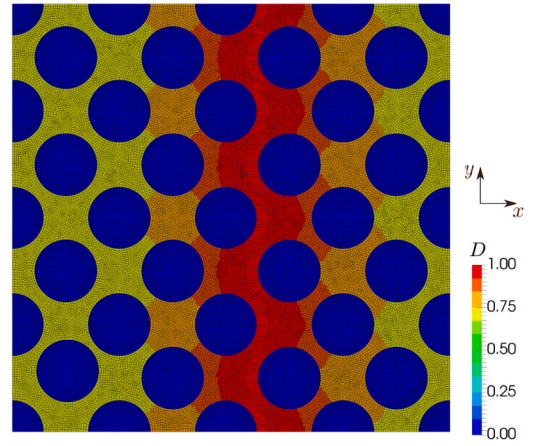
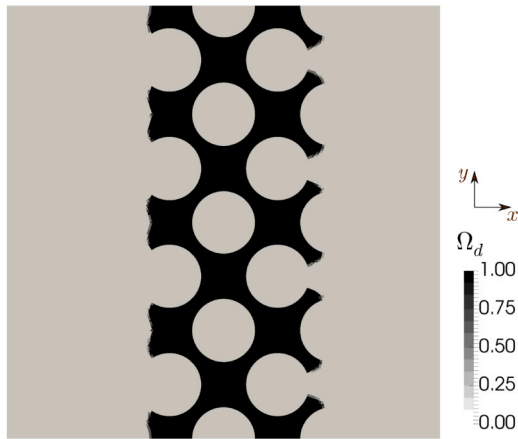
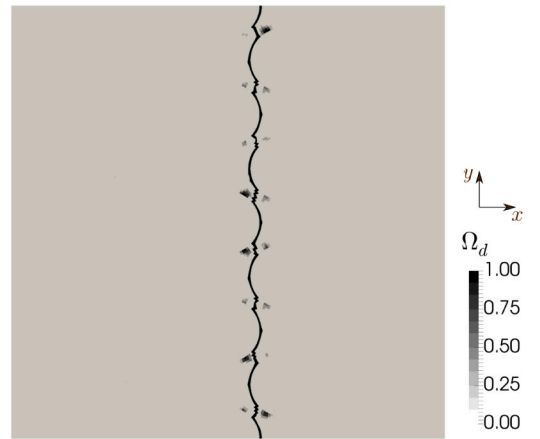
(a) $\{F_{xx}\}_{\Omega_0} = 1.0055$ (b) $\{F_{xx}\}_{\Omega_0} = 1.0155$ (c) $\{F_{xx}\}_{\Omega_0} = 1.0164$ (d) $\{F_{xx}\}_{\Omega_0} = 1.025$ (e) $\{F_{xx}\}_{\Omega_0} = 1.0164$ (f) $\{F_{xx}\}_{\Omega_0} = 1.025$

Fig. 11. Damage contour plots with mesh deformation factor 1 (Figs. a to d) and actively damaging zone Ω_d (Figs. e and f) for mesh size $0.25\mu\text{m}$ at different stages of deformation as indicated in Fig. 10.

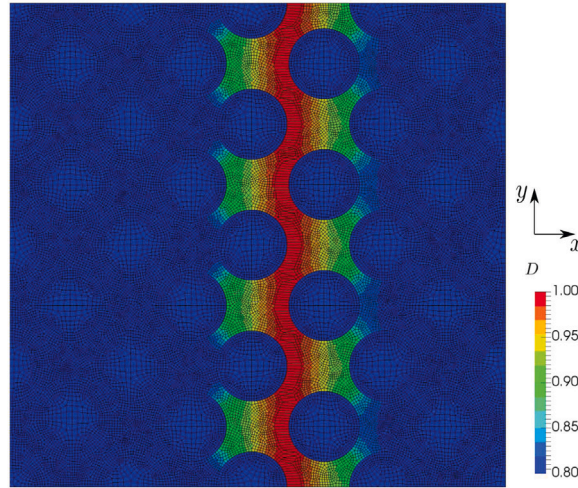


Fig. 12. Damage contour plot of the failure zone (i.e. volume of high damage) with mesh deformation factor 1 for mesh size $0.25\ \mu\text{m}$ at the end of deformation ($\{F_{xx}\}_{\Omega_0} = 1.025$) as indicated in Fig. 10.

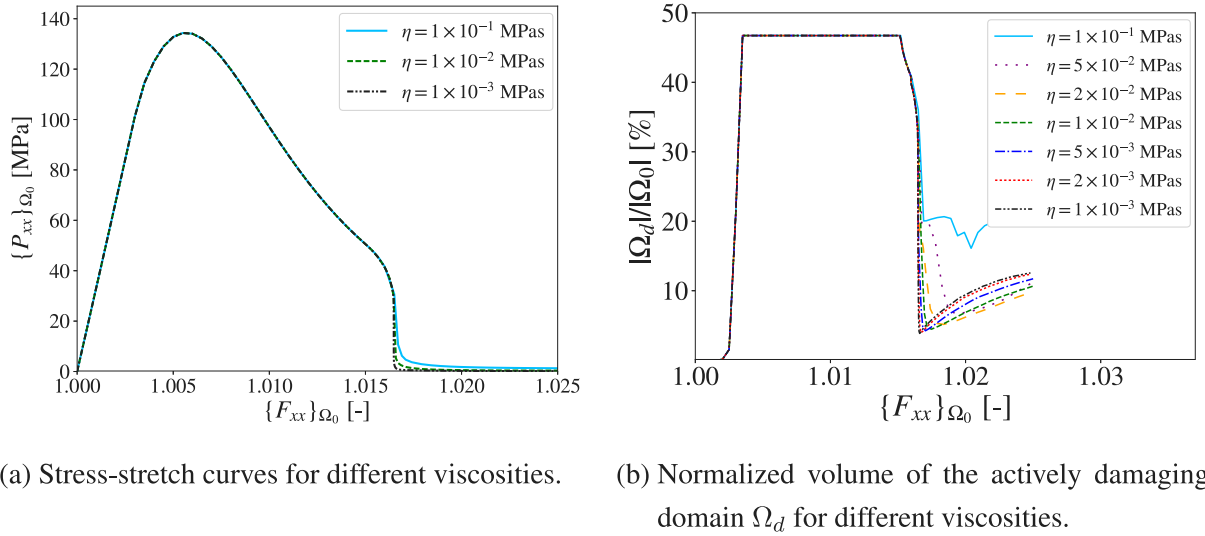


Fig. 13. Numerical results for different values of the pseudo viscosity for a regular RUC of size $48\ \mu\text{m} \times 48\ \mu\text{m} \times 1\ \mu\text{m}$.

6.3. RUC size comparison

A size convergence study was performed using five RUCs with regular pattern and varying sizes as shown in Fig. 6. Stresses and stretches were calculated using the volume average of (i) the complete computational domain Ω_0 following the standard homogenization approach and (ii) the actively damaging domain Ω_d according to failure zone homogenization scheme proposed here.

Stress–stretch curves based on standard homogenization are visualized in Fig. 14. The curves are almost identical in the elastic region. At the onset of strain softening (peak of the curve), the stress–stretch curve of the smallest RUC differs from the others. This effect results from the bigger influence of the imperfection on the smallest RUC. Thereafter, the curves differ in the onset of final failure, which is related to the formation of a localization band. Moreover, increasing the RUC size results in lower deformations at which the final load drop occurs.

In Fig. 15, the corresponding stress–stretch curves of the failure zone homogenization scheme are shown. As can be seen, the curves are independent of the RUC size. The small differences in the final (residual) stresses can be related to artificial viscosity. Due to the loading rate of the RUC being the same for all RUC sizes, the strain rate of the localization zone (and therefore the viscous residual stresses) slightly increase for increasing RUC size.

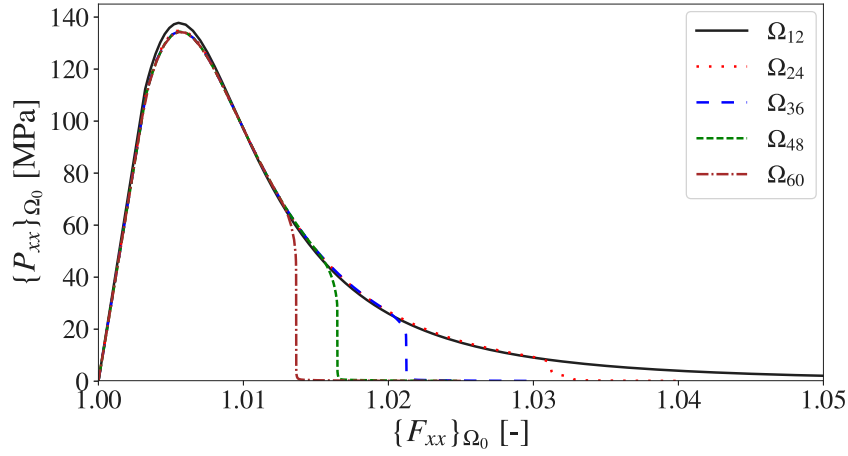


Fig. 14. Stress–stretch curves for different RUC sizes for standard homogenization. The RUC sizes range from $12 \mu\text{m} \times 12 \mu\text{m} \times 1 \mu\text{m}$ (Ω_{12}) to $60 \mu\text{m} \times 60 \mu\text{m} \times 1 \mu\text{m}$ (Ω_{60}).

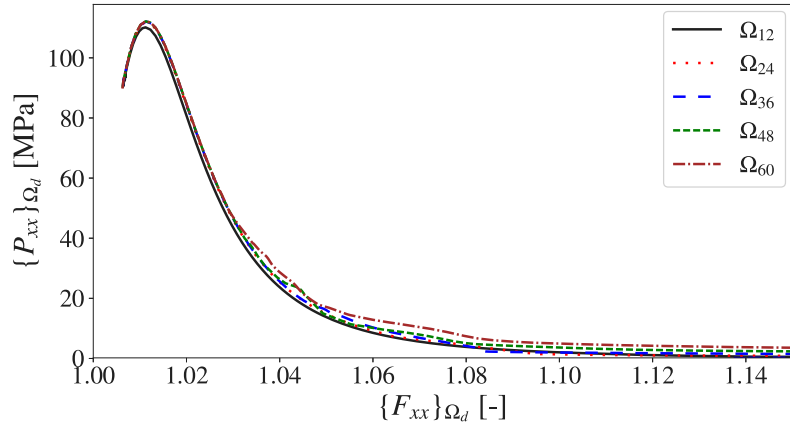


Fig. 15. Stress–stretch curves for different RUC sizes for failure zone homogenization. The RUC sizes range from $12 \mu\text{m} \times 12 \mu\text{m} \times 1 \mu\text{m}$ (Ω_{12}) to $60 \mu\text{m} \times 60 \mu\text{m} \times 1 \mu\text{m}$ (Ω_{60}).

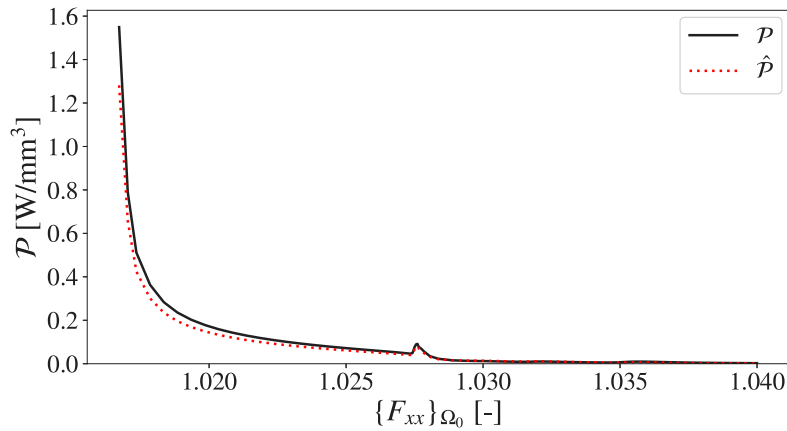


Fig. 16. Comparison of left and right hand side of Eq. (67) for the $48 \mu\text{m} \times 48 \mu\text{m} \times 1 \mu\text{m}$ RUC after onset of localization; average of the products ($P = \frac{1}{\Omega_0} \int_{\Omega_d} \mathbf{P}_m^T : \delta \mathbf{F}_m \, dV$, black) and product of the averages ($\hat{P} = \frac{1}{\Omega_0} \int_{\Omega_d} \mathbf{P}_m^T \, dV : \{\delta \mathbf{F}_m\}_{\Omega_d}$, red).

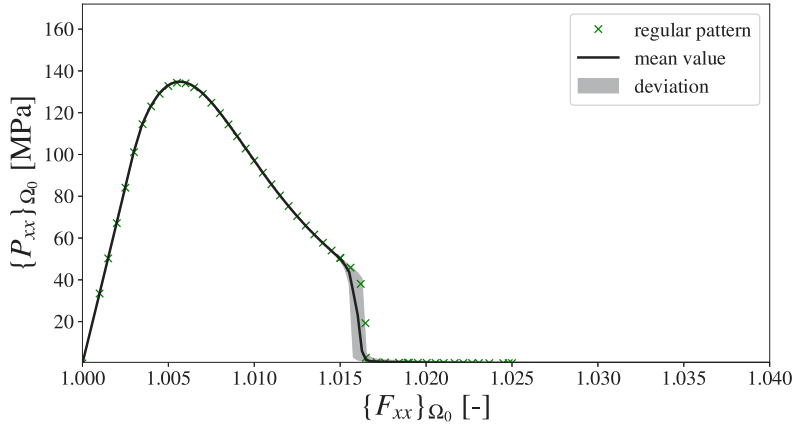


Fig. 17. Stress–stretch curves resulting from standard homogenization for *randomized* and regular RUCs of size $48\ \mu\text{m} \times 48\ \mu\text{m} \times 1\ \mu\text{m}$; black: Mean value of 10 realizations, gray: Scatter (maximum and minimum) of 10 realizations, green: Response of the regular RUC. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

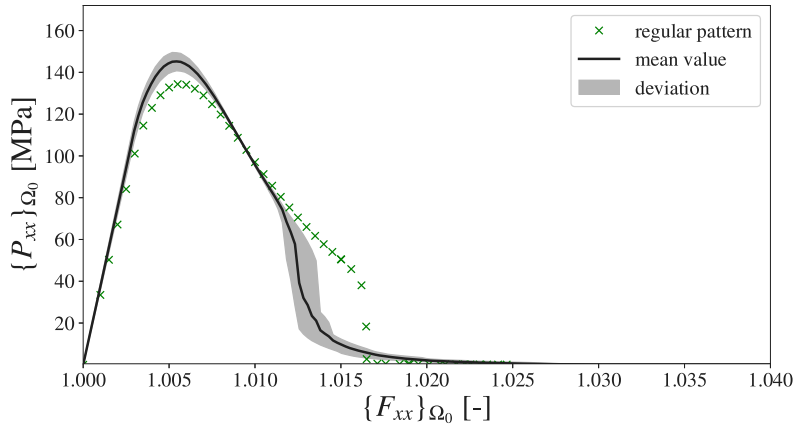


Fig. 18. Stress–stretch curves for RSA and regular RUCs of size $48\ \mu\text{m} \times 48\ \mu\text{m} \times 1\ \mu\text{m}$; black: Mean value of 10 realizations, gray: Scatter of 10 realizations, green: stress–stretch response of the regular RUC. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Numerical evidence for Eq. (67) is shown in Fig. 16. Here, the left and right hand side are given as individual curves. The curves are plotted from the onset of localization until end of simulation. Overall, the curves show a good correlation thus indicating that Eq. (67) holds. The small peak at $\{F_{xx}\}_{\Omega_0} = 1.028$ can be associated to viscous effects when the deformation localizes in on layer of elements as shown by Figs. 11(c) and 11(d).

6.4. Comparison of randomized and RSA RUCs

Two sets of 10 realizations each were created for the $48\ \mu\text{m} \times 48\ \mu\text{m} \times 1\ \mu\text{m}$ RUCs with the randomization approach and the RSA method, respectively. The corresponding stress–stretch curves homogenized over the whole domain Ω_0 are plotted in Figs. 17 and 18, respectively, together with the result of the regular RUC of the same size (green marks in Fig. 14). In what follows, the results are summarized by a mean curve and an encasing scatter showing the maximum and minimum of the ten realizations.

All ten realizations generated using the randomization approach (see Fig. 17) show the same global stress–stretch curves until failure. Here, the onset of failure differs due to differences in the microstructure. Generally, the curves for the regular RUC and the 10 *randomized* RUCs are close.

On the contrary, the stress–stretch curves of the RSA RUCs (see Fig. 18) differ in both, the peak load as well as the point of final failure. In addition, compared to *randomized* RUCs, all RSA RUCs show a higher peak load, higher elastic stiffness, lower stretches at failure, and the stress–stretch curves show a greater scatter. Interestingly, even though the maximum stress is different, the deformation at which the maximum stress is reached is quite the same for all realizations of both generation algorithms.

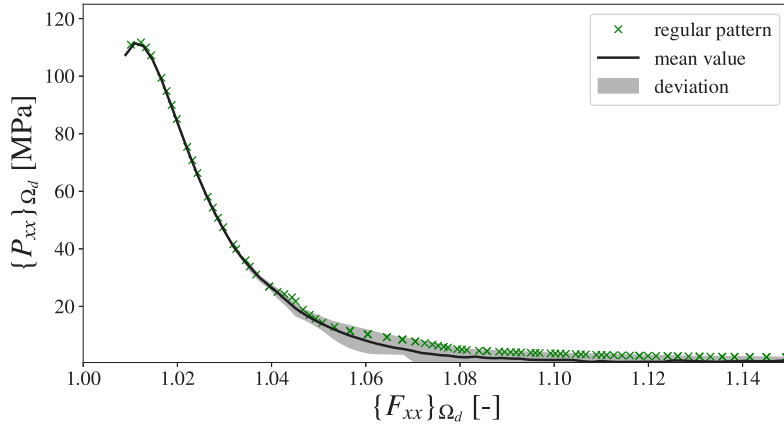


Fig. 19. Stress–stretch curves for failure zone averaging of *randomized* RUCs; black: Mean value of 10 realizations, gray: Scatter of 10 realizations, green: stress–stretch response of the regular RUC. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

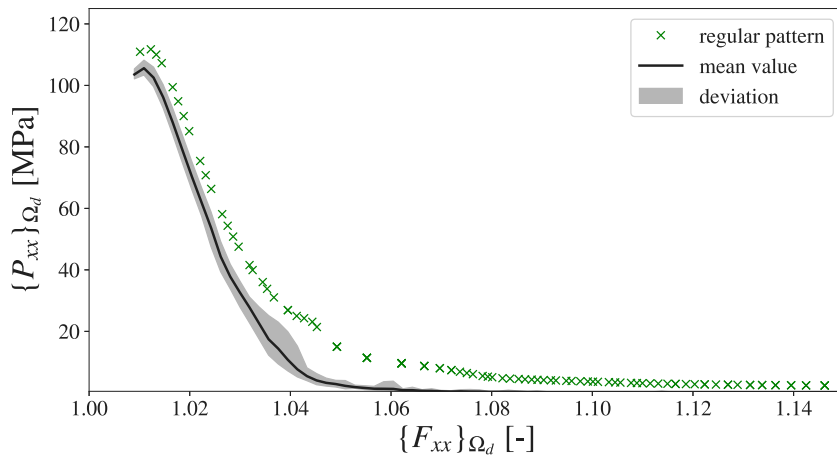


Fig. 20. Stress–stretch curves for failure zone averaging of *RSA* RUCs; black: mean value of 10 realizations, gray: upper and lower bound, green: stress–stretch response of the regular RUC. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Moreover, the failure zone homogenization scheme was applied to the same sets of realizations, such that the averaging was restricted to the actively damaging domain Ω_d . The corresponding stress–stretch relations are visualized in Figs. 19 and 20.

As illustrated in Fig. 19, the stress–stretch curves of the *randomized* RUC realizations do not show any differences until a deformation of 1.04 is reached. Thereafter, the curves diverge by a small amount until asymptotically converging towards zero stress. The stress–stretch curve of the regular pattern lies within the scatter (maximum and minimum) of the different curves of the ten realizations.

In contrast, the stress–stretch curves of the *RSA* RUCs show a noticeable scatter in the whole deformation range with a maximum scatter at about a stretch of 1.04. Overall, the peak stress is lower in comparison to the peak stress of the regular pattern. Additionally, the stress–stretch curves of the *RSA* RUCs approach zero stress at lower deformations than the curves of the regular or *randomized* RUCs.

All in all, the curves of the *RSA* RUCs show a higher deviation than the *randomized* RUCs. Failure appears earlier in terms of global stress–stretch curves and showed a much faster asymptotic approach towards zero stress when failure zone averaging is applied. While the peak of *RSA* curves is higher for standard averaging, it is lower when failure zone averaging is applied.

The effect of fiber clustering, present in the *RSA* approach, might explain the differences in the maximum peak load and elastic response of the *RSA* RUCs. In Kulosa et al. (2017), it was shown that in RUCs of sintered ceramics so-called load transfer paths develop. These load transfer paths are regions of high stresses which transfer most of the load through the RUC.

In Figs. 21(a) and 21(b), the first Piola–Kirchhoff stress contour plots of a *randomized* and a *RSA* RUC at the same far field deformation are shown. It shall be noted that the maximum stress of the *RSA* RUC was at approximately 50 MPa. For convenience and comparability, the color scale was lowered and chosen the same for both plots. Apparently, the stress distribution of the *RSA*

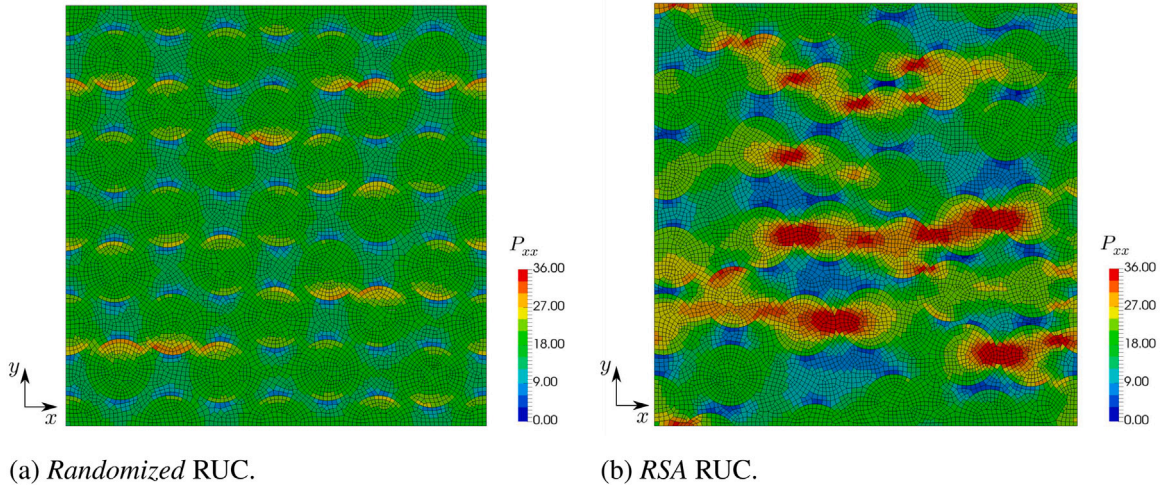


Fig. 21. First Piola-stress contour plots (in MPa) of a *randomized* and *RSA* RUC of size $48\mu\text{m} \times 48\mu\text{m} \times 1\mu\text{m}$ at an elastic far field deformation of $F_{xx} = 1.0005$.

RUC was much more heterogeneous and in line with the load transfer paths shown in Kulosa et al. (2017). These load paths mainly pass through fibers and close gaps between adjacent fibers. Overall, the fibers are loaded with higher stresses compared to the *Randomized* RUC in Fig. 21(a). These higher stresses in regions of higher stiffness are assumed to cause the difference in the elastic response.

The same effect is assumed to have caused the lower peak stress when failure zone averaging was applied. At the peak load, the actively damaging region was the matrix region. From Fig. 21(b) it can be seen that the matrix region shows small areas of high stresses and some larger areas of low or even zero stress. In an averaged sense, the regions of low stress outweighed the regions of high stresses, thus – in average – causing the lower peak stresses.

The results show that the fiber distribution has an impact on the elastic and inelastic response of the RUC. Therefore, the fiber distribution of the RUC should correspond to the fiber distribution found in CFRPs. Since the *RSA* method provides fiber distributions which are closer to the fiber distribution in real CFRPs (see e.g. Fig. 13 in París et al., 2018), this method will be used for RUC generation in the following.

6.5. Statistical investigation of representative RUC size

A statistical investigation of the RUC size ranging from an edge length of $12\mu\text{m}$ to $60\mu\text{m}$ was performed (cf. Fig. 6). For each RUC size, ten realizations were generated using the *RSA* method. First, stresses and stretches were calculated using the standard averaging over the entire domain Ω_0 . The resulting homogenized stress–stretch curves are summarized in Figs. 22 to 25. Further, the results for RUCs with an edge length of $48\mu\text{m}$ have been displayed already in Fig. 18.

In Figs. 22 to 25 one can observe that the stress–stretch curves show less differences in the elastic regime and the peak stresses, the bigger the RUC size gets. Also, larger RUCs show an earlier load drop, which agrees with the findings from Section 6.3. Moreover, when comparing the results of *RSA* RUCs to those with regular patterns, the peak stresses of the *RSA* RUCs are larger and the load drops take place at lower stretches.

Additionally, in order to get more insight into the question of representativeness, the mechanical work density of the entire domain Ω_0 was computed according to Eq. (71). The result is plotted in Fig. 26 together with the corresponding deviation. In addition to the results generated from the *RSA* RUCs, the results from the *regular* RUCs (including new RUC sizes of $72\mu\text{m}$ and $84\mu\text{m}$) are shown here as well. It can be concluded, that the variation decreases with larger RUC size, but the mean values do not show any convergence with respect to the RUC size. Hence, no representative RUC size can be identified using the standard homogenization approach.

Furthermore, the same RUCs also were investigated using the failure zone homogenization scheme with averaging over the actively damaging domain Ω_d , leading to the stress–stretch curves shown in Figs. 27 to 30.

Figs. 27 to 30 confirm the same trends already explained in the previous Section 6.4. When failure zone homogenization is used, the maximum stresses of the *RSA* RUCs generally are lower compared to the maximum stresses of the *regular* RUCs. However, in Fig. 27 the upper bound of the $12\mu\text{m}$ *RSA* RUCs does exceed the peak stress of the *regular* RUC of the same size. Nevertheless, even for this RUC size, the mean of the maximum stresses is lower. Noteworthy, the difference between the *regular* and *RSA* RUCs increases with increasing RUC size. Therefore, it is concluded that geometric features do play an important role in strain hardening as well as strain softening.

Concluding, also the mechanical work density of the failure zone Ω_d was investigated. The result is plotted in Fig. 31 together with the corresponding deviation. In addition to the results generated from the *RSA* RUCs, the results from the *regular* RUCs

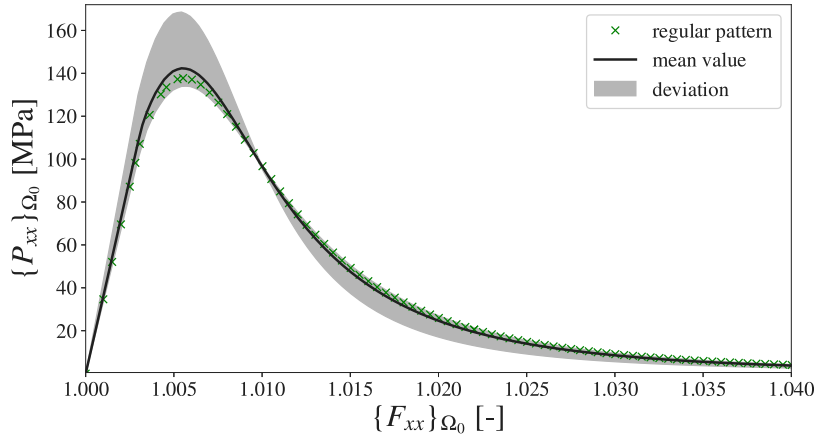


Fig. 22. Stress–stretch curves resulting from standard homogenization for RSA and regular RUCs of size $12\,\mu\text{m} \times 12\,\mu\text{m} \times 1\,\mu\text{m}$; black: Mean value of 10 realizations, gray: Scatter of 10 realizations, green: Response of the regular RUC. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

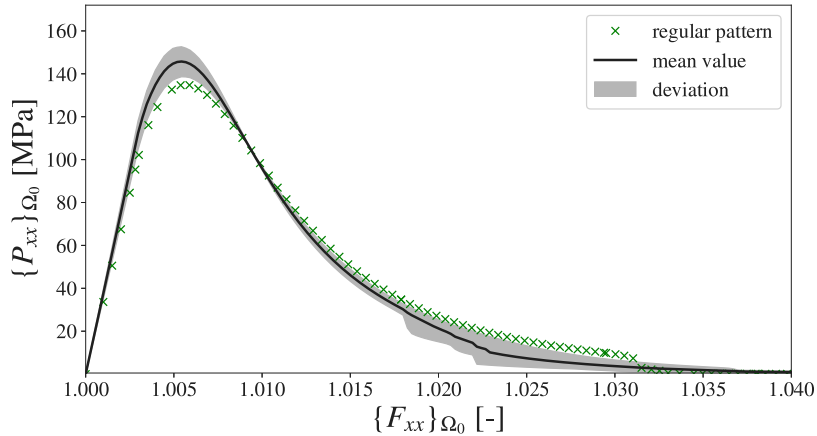


Fig. 23. Stress–stretch curves resulting from standard homogenization for RSA and regular RUCs of size $24\,\mu\text{m} \times 24\,\mu\text{m} \times 1\,\mu\text{m}$; black: Mean value of 10 realizations, gray: Scatter of 10 realizations, green: Response of the regular RUC. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(including new RUC sizes of $72\,\mu\text{m}$ and $84\,\mu\text{m}$) are shown here as well. Similarly to the standard homogenization case (see Fig. 26), the overall higher variation decreases with increasing RUC size. But in contrast to the averaged global mechanical work density for the entire domain, the mean value decreases way less when the failure zone homogenization scheme is applied. Interestingly, the results of the regular RUCs and the RSA RUCs coincide better. However, when regarding the *regular* RUC size of $84\,\mu\text{m}$, an increase in dissipation is observed. This is due to higher strain rates within the failure zone when the RUC size is increased. This leads to considerable viscous stresses from the pseudo viscosity and in turn higher dissipated energy. However, applying the proposed failure zone homogenization approach results in a much lower slope of the dissipated work density. As such, an RUC size of $72\,\mu\text{m}$ is considered converged.

6.6. Numerical investigation of micromorphic power

In Sections 4.1 and 4.3, it was shown analytically that the micromorphic virtual work has to vanish when integrated over the entire volume, but not necessarily when failure zone averaging is applied. In order to investigate the effect of this micromorphic virtual work in the failure zone, a parameter study of the micromorphic parameters l_1 and H^χ was performed in Poggenpohl and Simon (2021). There, an RUC of size $60\,\mu\text{m} \times 60\,\mu\text{m} \times 1\,\mu\text{m}$ with regular pattern was evaluated w.r.t. the influence of these parameters on the micromorphic power integrated over the whole computational domain and the actively damaging zone, respectively. Exemplarily, one result from this investigation with a specific set of micromorphic parameters ($l_1 = 3.2 \times 10^{-4}\,\mu\text{m}$ and $H^\chi =$

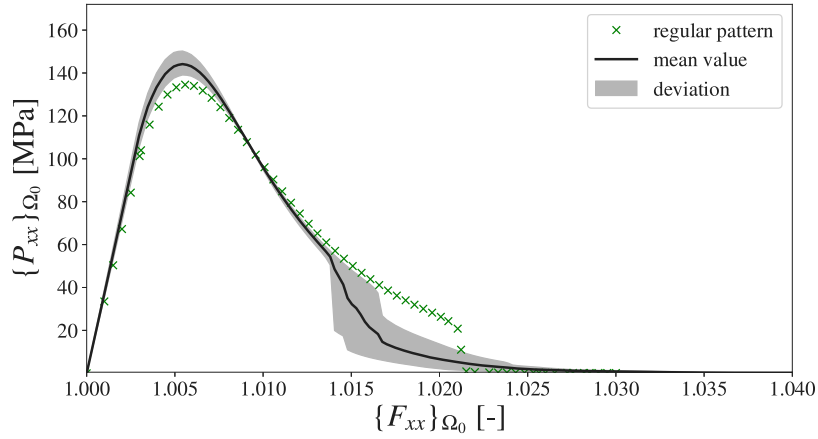


Fig. 24. Stress–stretch curves resulting from standard homogenization for RSA and regular RUCs of size $36\ \mu\text{m} \times 36\ \mu\text{m} \times 1\ \mu\text{m}$; black: Mean value of 10 realizations, gray: Scatter of 10 realizations, green: Response of the regular RUC. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

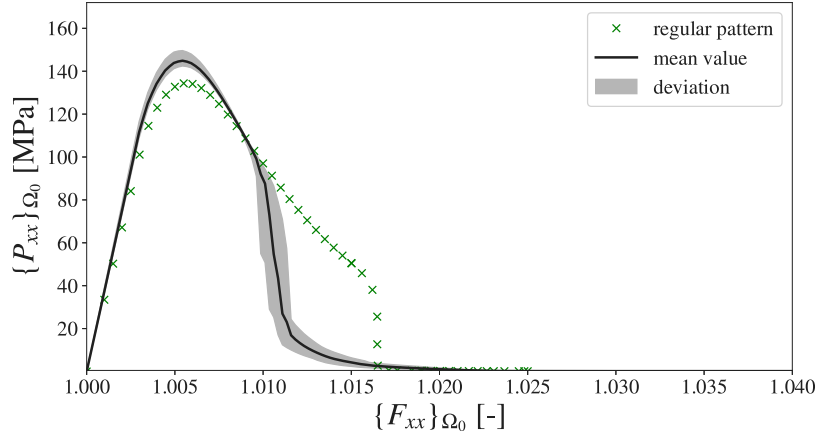


Fig. 25. Stress–stretch curves resulting from standard homogenization for RSA and regular RUCs of size $60\ \mu\text{m} \times 60\ \mu\text{m} \times 1\ \mu\text{m}$; black: Mean value of 10 realizations, gray: Scatter of 10 realizations, green: Response of the regular RUC. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

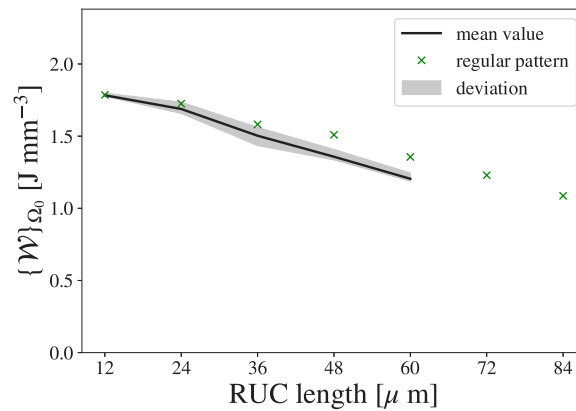


Fig. 26. Summary of the total mechanical work density of the simulated RSA RUCs.

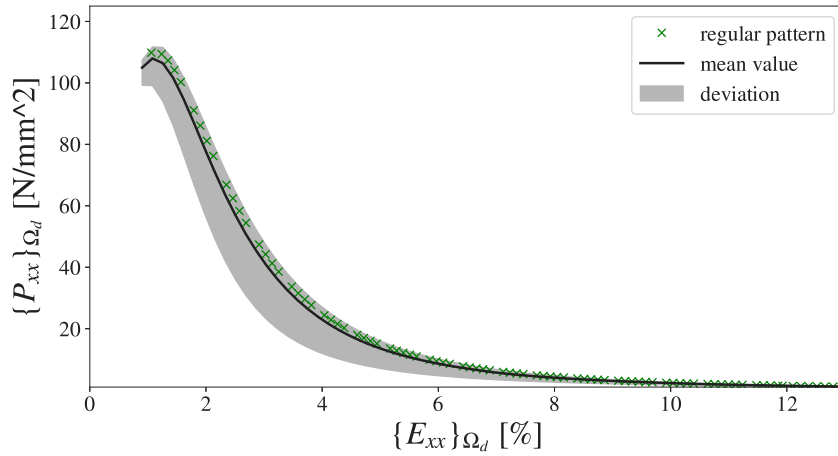


Fig. 27. Stress–stretch curves resulting from failure zone homogenization for RSA and regular RUCs of size $12\mu\text{m} \times 12\mu\text{m} \times 1\mu\text{m}$; black: Mean value of 10 realizations, gray: Scatter of 10 realizations, green: Response of the regular RUC. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

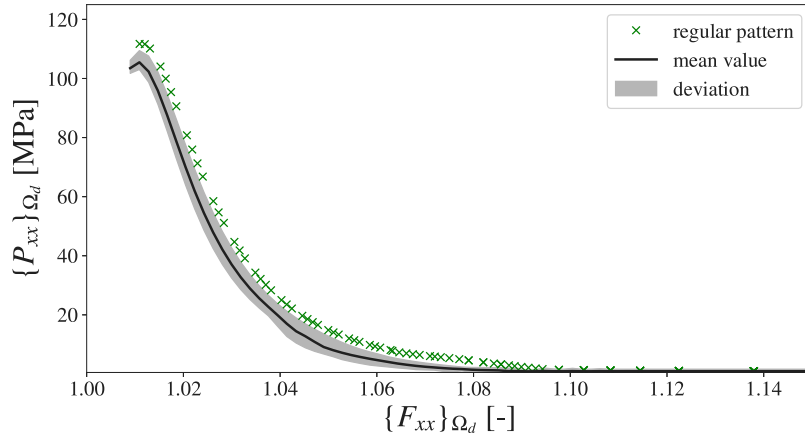


Fig. 28. Stress–stretch curves resulting from failure zone homogenization for RSA and regular RUCs of size $24\mu\text{m} \times 24\mu\text{m} \times 1\mu\text{m}$; black: Mean value of 10 realizations, gray: Scatter of 10 realizations, green: Response of the regular RUC. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.0×10^2 MPa) is shown in Fig. 32. Here, the term power density refers to the time derivative of the mechanical or micromorphic work density, i.e.

$$\mathcal{P}_{mech} = \mathbf{P} : \dot{\mathbf{F}} \quad (73)$$

$$\mathcal{P}_{mic} = \mathbf{b} \cdot \nabla \dot{\mathbf{D}} + a \dot{\mathbf{D}}. \quad (74)$$

As can be observed in Fig. 32, the micromorphic power density integrated over the whole computational domain Ω_0 is negligible compared to the mechanical power density, which numerically substantiates the analytical result of Eq. (49).

In contrast, the micromorphic power density does have significant values during the formation of the localized failure zone, i.e. during the load drops, when only the actively damaging domain Ω_d is considered (see Fig. 33). Thus, this result confirms the findings in Section 4.3 noting that the micromorphic power (and therefore power density) of the failure zone does not necessarily vanish, even though it is zero before and shortly after the load drop (cf. zoom in of Fig. 33). It shall be noted that for better comparison, the micromorphic power density of the actively damaging domain was scaled to the whole computational domain, i.e.

$$\mathcal{P}_{\chi,d} = \frac{\Omega_d}{\Omega_0} \{ \mathcal{P}_{\chi} \}_{\Omega_d}. \quad (75)$$

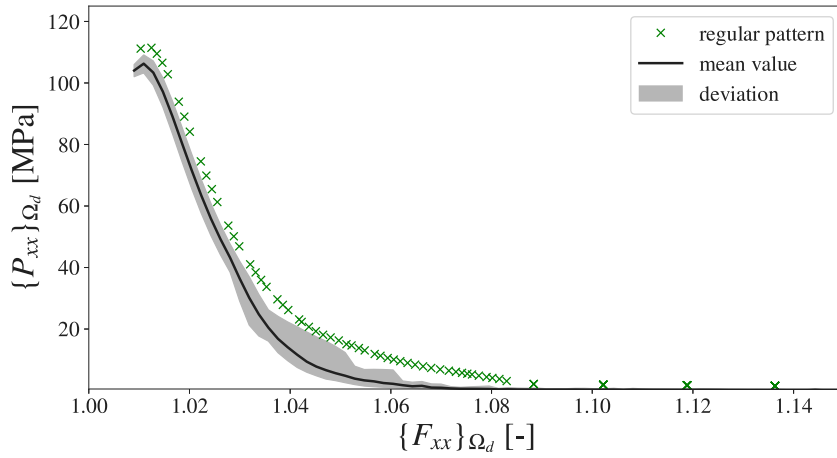


Fig. 29. Stress–stretch curves resulting from failure zone homogenization for RSA and regular RUCs of size $36\mu\text{m} \times 36\mu\text{m} \times 1\mu\text{m}$; black: Mean value of 10 realizations, gray: Scatter of 10 realizations, green: Response of the regular RUC. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

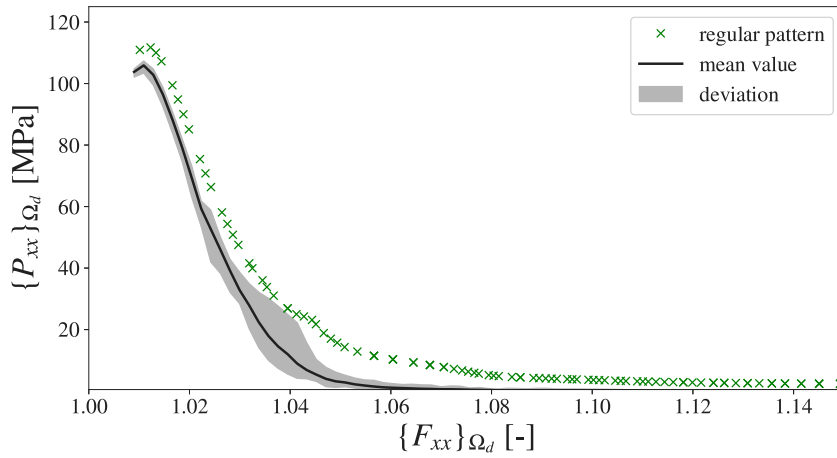


Fig. 30. Stress–stretch curves resulting from failure zone homogenization for RSA and regular RUCs of size $60\mu\text{m} \times 60\mu\text{m} \times 1\mu\text{m}$; black: Mean value of 10 realizations, gray: Scatter of 10 realizations, green: Response of the regular RUC. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

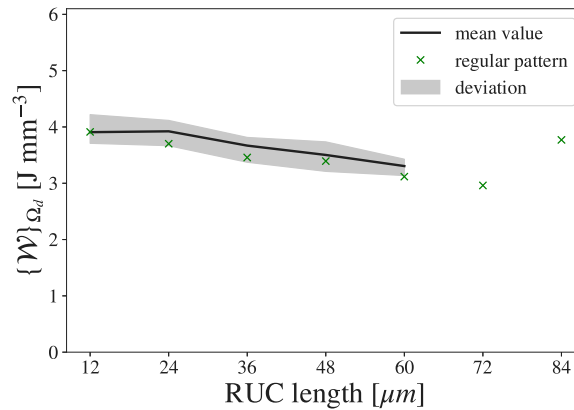


Fig. 31. Summary of the mechanical work density of the failure zone of simulated RSA RUCs.

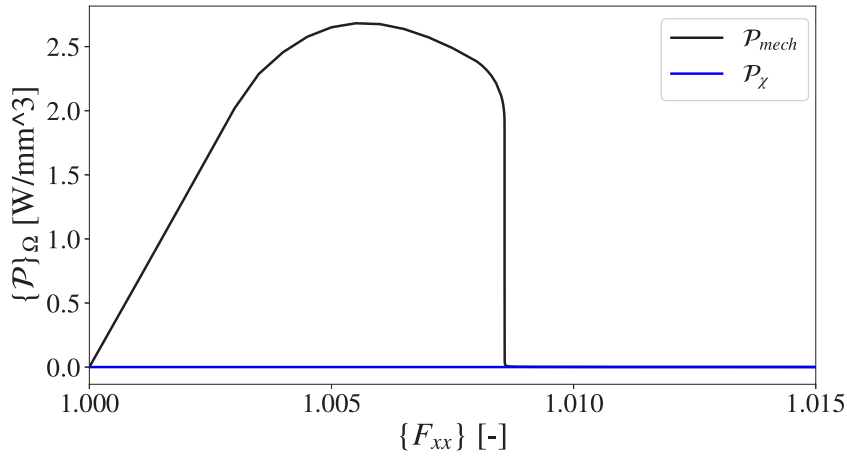


Fig. 32. Mechanical and micromorphic power averaged over the whole computational domain. Source: (Poggenpohl and Simon, 2021).

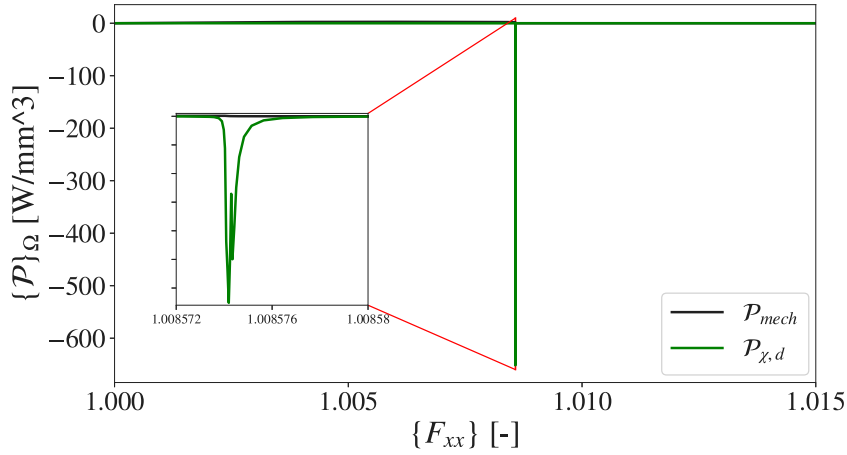


Fig. 33. Mechanical P_{mech} and micromorphic power $P_{\chi,d}$ of the actively damaging region Ω_d , both averaged over the whole computational domain Ω_0 . Source: (Poggenpohl and Simon, 2021).

Table 3

Comparison of the work densities of RSA RUCs with different RUC sizes; mean values from 10 realizations each.

	12 μm	24 μm	36 μm	48 μm	60 μm
$\mathcal{W}_{mech}$ [J mm ⁻³]	1.7774	1.6816	1.4977	1.3471	1.1859
$\mathcal{W}_{\chi,d}$ [J mm ⁻³]	-0.0175	-0.0163	-0.0141	-0.0118	-0.0099
$\mathcal{W}_{\chi,d}/\mathcal{W}_{mech}$ [%]	-0.9865	-0.9709	-0.9416	-0.8769	-0.8323

Finally, the mean mechanical work density $\mathcal{W}_{mech}$ and the mean micromorphic work density $\mathcal{W}_{\chi,d}$ of the actively damaging domain were compared for the RSA RUCs from Section 6.5 with different sizes and 10 realizations each. Both quantities were calculated as time integrals of the corresponding power densities P_{mech} and $P_{\chi,d}$, respectively. The results are presented in Table 3.

As can be seen from Table 3, the overall work generated by the micromorphic terms of the actively damaging domain is negative. It is much lower (in terms of absolute values) than the overall work generated by the mechanical terms as well. Findings in Poggenpohl and Simon (2021) showed that the micromorphic power only took significant values for a short period of time and were close to zero otherwise (cf. Fig. 33). Thus, low values for the overall generated work in combination with much higher peaks are plausible. Overall, the quotient of $\mathcal{W}_{\chi,d}$ and $\mathcal{W}_{mech}$ is lower than 1% in the means of absolute values. For the parameter set chosen, the overall micromorphic power appears to be of minor importance. However, it is shown that the micromorphic power can take significant values and therefore should not be neglected without being proven negligible.

Table 4
Material parameters of the interface.

Parameter	Value	Parameter	Value	Parameter	Value
t_0	50 MPa	λ_0	$2 \times 10^{-2} \mu\text{m}$	λ_f	$2 \times 10^{-1} \mu\text{m}$
η_{CZ}	0.5 MPa s m^{-1}	β	1	H_{pen}	5000 MPa mm^{-1}

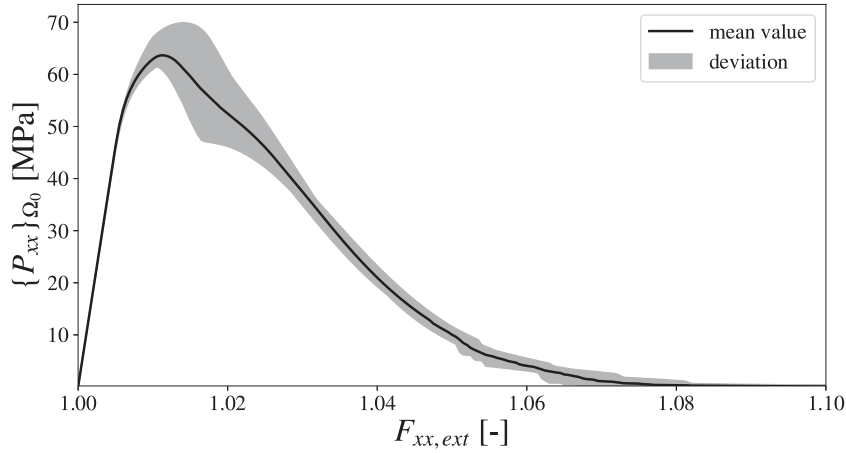


Fig. 34. Stress-stretch curves resulting from standard homogenization for RSA RUCs of size $24 \mu\text{m} \times 24 \mu\text{m} \times 1 \mu\text{m}$ with damaging interfaces; black: Mean value of 10 realizations, gray: Scatter of 10 realizations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

6.7. Results of RSA RUCs with damaging interfaces

Ten realizations of size $24 \mu\text{m} \times 24 \mu\text{m} \times 1 \mu\text{m}$ were simulated accounting for both, damaged and debonding induced softening. This reduced RUC size was chosen despite the question of representative size (c.f. Section 6.5). Reason for this are the numerically expensive simulations when bulk damage and debonding are simulated.¹ To model the debonding of interfaces between fiber and matrix, the cohesive zone elements introduced in Section 3.2 were used. The material properties of the matrix and fibers of Table 1 were retained, and the thickness of the interface was chosen to be of $1/1000$ of the fiber diameter (i.e. $7 \times 10^{-3} \mu\text{m}$).

The material properties of the interfaces are summarized in Table 4, where t_0 denotes the maximum traction of the interface. This value was taken from experimental investigations of carbon fibers–epoxy matrix interfaces presented in Zhang et al. (2018). The cohesive zone gap opening at maximum traction (λ_0) and at full failure (λ_f) were chosen to give reasonable results. Further, β was chosen to be one such that the shear and the normal gap opening contributed to the effective gap opening by the same factor. As for the bulk material model, a pseudo viscosity (η_{CZ}) was used. This parameter was chosen as small as possible while achieving numerical stability. Self penetration was avoided with help of the penalty parameter H_{pen} , for which half the stiffness of the weakest bulk constituent was used as suggested e.g. in Rezaei et al. (2017).

In Fig. 34, the stresses for ten realizations using standard homogenization are summarized. As shown in Section 4.1, the homogenization of the deformation gradient at the presence of damaging interfaces is not straight forward. Therefore, the externally applied (macroscopic) deformation gradient is used here. It can be observed, that the peak stress of RUCs with damaging interfaces is much lower compared to the results shown in Fig. 23 without debonding. Additionally, the deviation of the different realizations is higher at the beginning of softening. Starting at $F_{xx,ext} = 1.03$, the deviation is reduced and comparable to the deviation of RUCs with perfect interfaces. At about $F_{xx,ext} = 1.08$, the stresses are zero, indicating complete failure. The far field deformation at the peak load varies as well. This is in contrast to the RUCs with perfect interfaces, where the stretch at peak load was the same for the same RUC size.

The work densities of the different constituents are summarized in Fig. 35. Here, $\mathcal{W}_{ext}$ denotes the external work density of the RUC, while $\mathcal{W}_{\Omega_d}$ and $\mathcal{W}_{\Omega_e}$ denote the work densities of the active damaging domain and the elastic domain of the bulk, respectively. The active damaging and elastic work densities of the interface are denoted by $\mathcal{W}_{\Gamma_d}$ and $\mathcal{W}_{\Gamma_e}$, respectively. Noteworthy, since the quantities shown here are work related quantities, negative values are possible. This is especially the case for material points outside of the localization zone. During damage loading (positive contribution) these material points contribute to the damaging work densities, while at elastic unloading (negative contribution) they contribute to the elastic work densities. In Figs. 36(a) to 36(c), three damage contour plots showing the deformed mesh are visualized. Their corresponding solution times are indicated in Fig. 35.

¹ The simulations took about 15 to 20 days on Intel Xeon E5-2670 and E5-2667 v2 processors using 2 cpus per simulation.

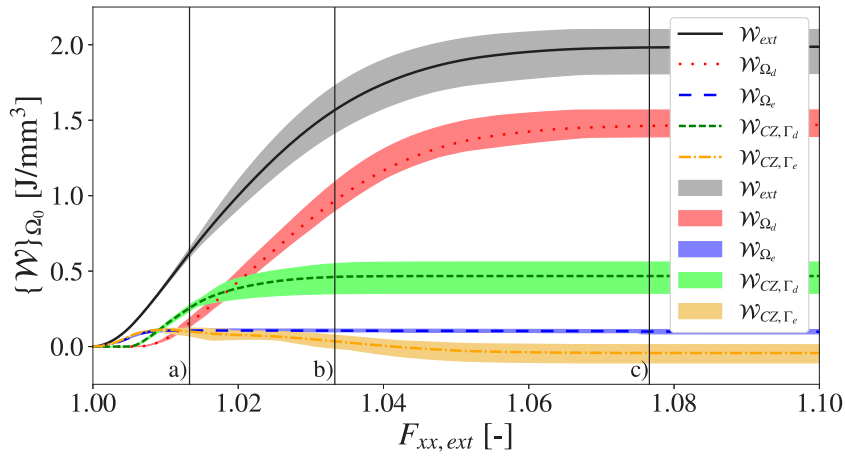


Fig. 35. Work contributions of the different constituents for the 24 μm RSA RUCs with damaging interfaces; black: external work, red: actively damaging matrix material, blue: elastic matrix material, green: actively damaging interface and orange: elastic interface; damage contour plots of one realization showing mesh deformation of different stages a) to c) are shown in Figs. 36(a) to 36(c). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

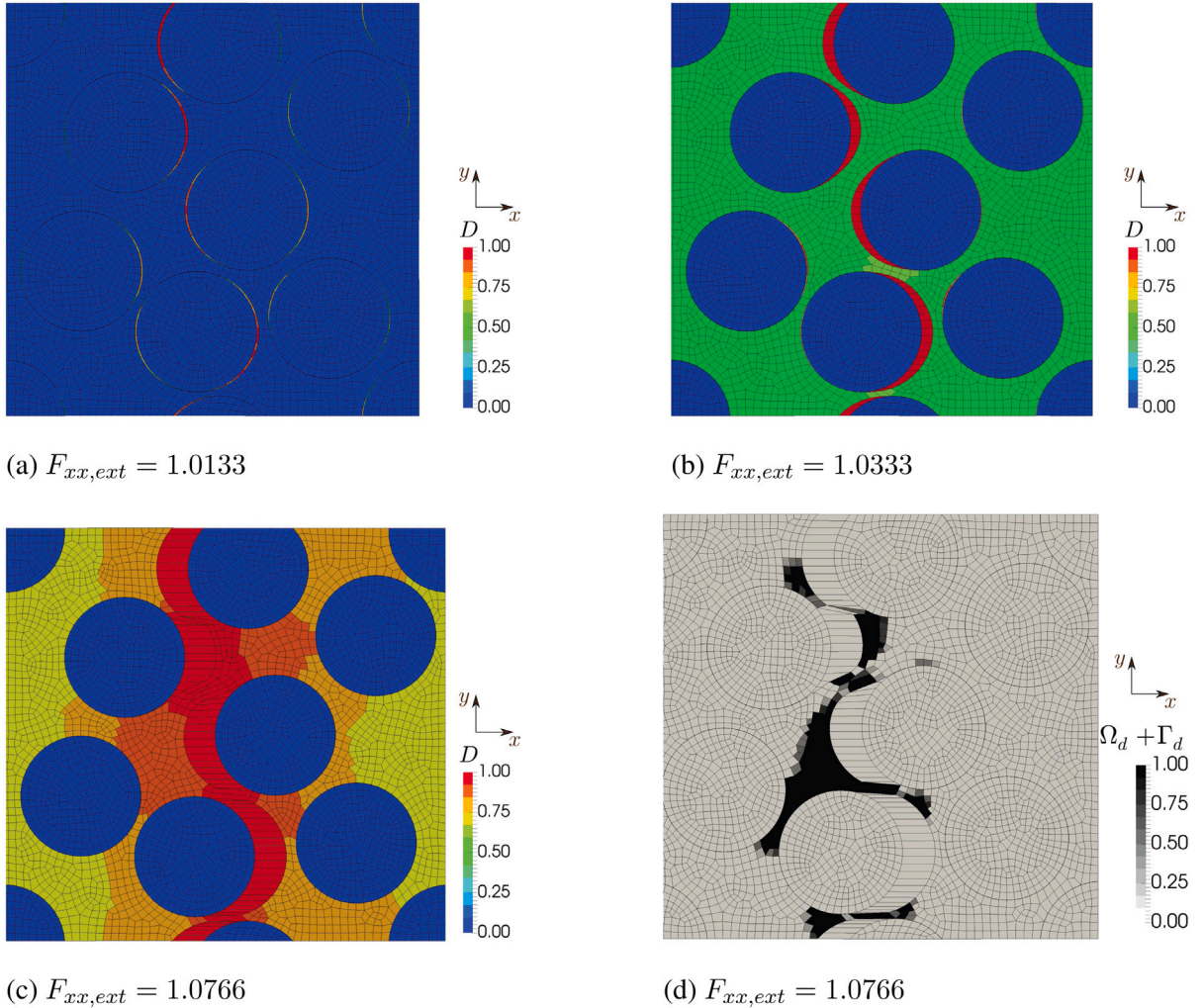


Fig. 36. Damage contour plots (a to c) and actively damaging zone for bulk and cohesive zone elements at different stages of deformation as indicated in Fig. 35 with mesh deformation factor 1.

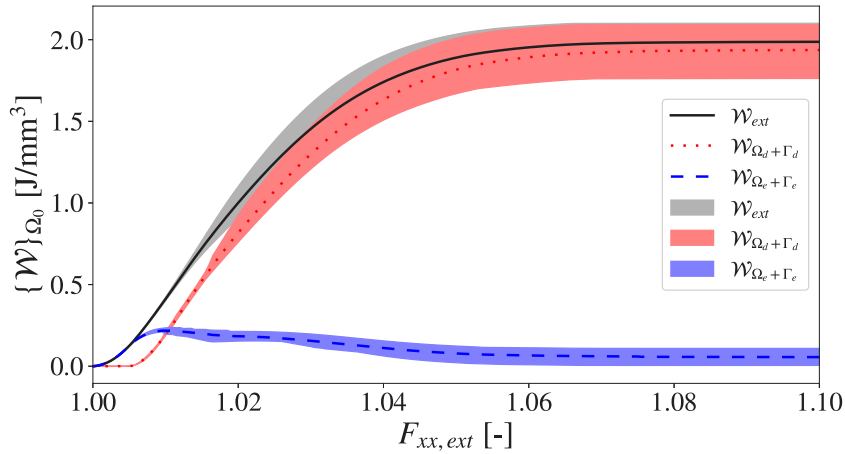


Fig. 37. Work contributions for the 24 μm RSA RUCs with damaging interfaces; red: actively damaging constituents, blue: elastic constituents and black: external work. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

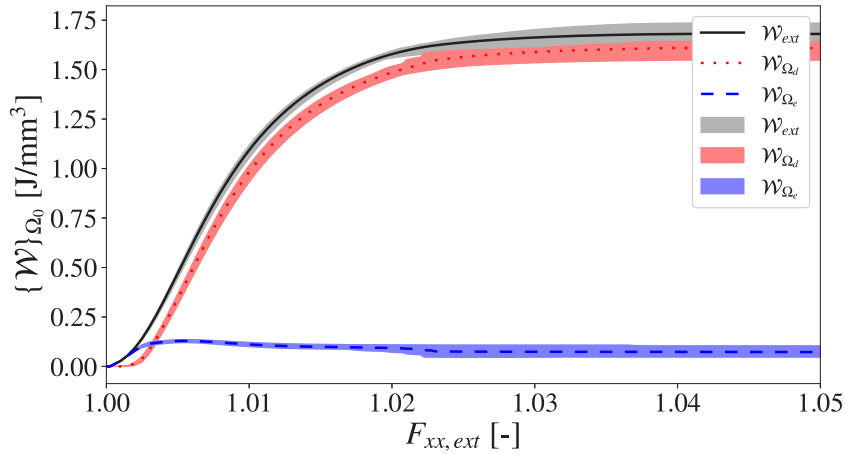


Fig. 38. Work contributions for the 24 μm RSA RUCs without damaging interfaces from previous Section 6.5; red: actively damaging constituents, blue: elastic constituents and black: external work. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The evolution of the virtual work densities shown in Fig. 35 is as follows. The external work first is governed by elastic parts of the bulk and the interface. Then, the interface start to damage and the work of both elastic parts stays constant. Subsequently, the bulk material begins to fail and the work of the elastic part of the interfaces decreases. At about $F_{xx,ext} = 1.02$, the work of the actively damaging bulk exceeds the work of the actively damaging interface, which stays constant from about $F_{xx,ext} = 1.03$ on. At about $F_{xx,ext} = 1.08$, all work quantities are constant indicating that the failure process is finished. This particular order of failure mechanisms was reported by Guillén-Hernández et al. (2020) as well, where the so called in-situ strength effect was investigated numerically. The failure mechanisms are shown in Figs. 36(a) to 36(c) as well. From Fig. 36(a), the opening of the interfaces is observed, where no significant damage is accumulated within the matrix. Fig. 36(b) shows the subsequent accumulation of damage within the matrix without the formation of a localization band. In Figs. 36(c) and 36(d), final failure with a localization zone is visualized. As can be seen, the crack mainly extends over the interphase between fiber and matrix while only the matrix shows an actively damaging zone. This is due to the interfaces being fully damaged with a complete loss of interface stiffness.

Further, the work densities of the overall actively damaging and elastic domain are compared to the external work in Fig. 37. Here, $\mathcal{W}_{\Omega_d + \Gamma_d}$ and $\mathcal{W}_{\Omega_e + \Gamma_e}$ describe the combined overall work densities of the actively damaging and elastic domains of bulk and interface, respectively.

Here, the external work is completely governed by the bulk until $F_{xx,ext} = 1.006$. At the onset of damage evolution at $F_{xx,ext} = 1.006$, the work in the bulk then slowly decreases, while the work of the actively damaging domain increases faster than the external work. At the end, the external work and the work of the actively damaging domain are almost equal while the work of the bulk is close to zero.

For comparison, the same work densities as in Fig. 37 are shown in Fig. 38 with perfect interfaces. Both curves show a similar behavior. Until the onset of failure, the external work is completely governed by the elastic part. Thereafter, the work of the actively

damaging domain and the external work evolve identically while the work of the elastic part stays constant. Finally, the external work is nearly completely governed by the work of the actively damaging domain.

While qualitatively showing the same effects, differences between the simulations with and without damaging interfaces arose from the computational time and the deviations. When debonding of interfaces was included, the RUCs failed at far field deformations that were twice as large as in the case with perfect bonding. Moreover, the external work density was higher for the RUCs with damaging interfaces, even though the peak stresses were significantly lower. It is concluded that the higher work density is a result of the much higher deformation. Finally, the RUCs with damaging interfaces showed a higher scattering compared to the RUCs with perfect interfaces.

7. Conclusion and outlook

In this work, a failure zone homogenization scheme was proposed based on the approach of [Nguyen et al. \(2010\)](#). The aim was to develop a homogenization strategy which takes into account all energetic contributions and finite strains. For this, the newly developed failure zone homogenization approach was tested using a statistical analysis of RUCs of carbon fiber reinforced plastics. For the simulation of material failure at the micro-scale, a St. Venant type material model for finite strains with isotropic damage was introduced. Further, to overcome undesired mesh dependence, a gradient extension based on the micromorphic approach is used. Additionally, the debonding of interface between carbon fibers and epoxy matrix was simulated using cohesive zone elements.

A possible extension of the material model used would be the introduction of plasticity and damage via a so-called two-surface material model as shown, e.g. in [Brepols et al. \(2017\)](#) or [Holthusen et al. \(2020\)](#). However, the experimental data provided by [Poggenpohl et al. \(2020\)](#) did not show considerable plastic deformations. Therefore, the material model used was restricted to brittle damage.

To ensure that no numerical artifacts affect the statistical analysis and that the homogenization procedures work, the following tests were performed: (I) A mesh convergence study, (II) a sensitivity analysis for the artificial viscosity, (III) a size convergence study as well as (IV) an analysis of different RUC generation algorithms. From the mesh convergence study it was shown that the mesh dependency was successfully cured. Further, a sensitivity analysis showed that introducing an artificial viscosity for stabilization had no effect on the overall stress–stretch response, while a significant effect on the evolving actively damaging domain was observed.

Further, the RUC size convergence study showed that standard averaging based on Hill's approach showed satisfactory results in the elastic region but RUC size dependent results in the softening region. In contrast, the failure zone homogenization scheme showed converged results w.r.t. the RUC size even in the softening region. Finally, two different RUC generation algorithms *randomized* and *RSA* were compared. It was shown that the different generation algorithms led to significant differences in the overall as well as the failure zone stress–stretch curves. Since the *RSA* RUCs provided fiber distributions which were closer to the distributions of real CFRPs, this generation method was used for the statistical analysis.

For the statistical analysis, five different RUC sizes with ten realizations each were evaluated. It was shown that Hill's approach led to converging results until maximum stress. However, the stress–stretch curves showed failure at lower stretch with increasing RUC size. Analysis of the dissipated work density supported these results.

Statistical analysis using the failure zone averaging showed convergence with respect to the RUC size which was supported by an analysis of the dissipated work density as well. Additionally, an analysis of the micromorphic power and work densities of the failure zone averaging of the *RSA* RUCs was conducted. It was shown that the micromorphic power density of the failure zone gives significant values during the formation of localization while the overall work density is small. Thus, it was concluded that for accurate failure zone homogenization, the contributions of micromorphic terms have to be considered. It is assumed that other gradient enhanced methods show a similar contribution of the power and work density. Thus, these contributions should be addressed when using failure zone averaging.

Finally, a statistical analysis of RUCs with damaging interfaces was conducted. In total, ten realizations of one particular RUC size were used. Stress strain curves show a lower peak stress and a higher scatter in the stress response. The use of bulk and cohesive zone elements allowed for an analysis of the energetic contributions of both softening effects. It was shown that the energetic contribution of the interface is lower than that of the bulk but still significant. Additionally, the RUCs with non perfect interfaces showed a higher dissipation potential and doubled far field strains until final failure.

In brief, the argument given in [Nguyen et al. \(2010\)](#) still holds: if the homogenization is performed within the failure zone, an RVE can be uniquely identified. In the current work, this concept was underpinned through deriving and evaluating the corresponding energetic contributions analytically and numerically. It was shown that the RVE still exists even for large deformations, with micromorphic/gradient-extended damage formulations, and when both, damage- and debonding-induced softening was considered. However, when homogenizing stresses and strains within a multiscale framework (out of the scope of this paper) the energy contributions from the boundary terms of the damaging domain must be considered.

In future works, the new homogenization approach shall be applied to shear and mixed mode loading. Further, the approach could – in principle – be applied to FE^2 simulations as shown in [Nguyen et al. \(2012\)](#), where a traction separation law is defined by micro simulations. However, the stress and stretch within the failure zone are not defined a priori, but calculated a posteriori. Thus, new or updated boundary conditions would need to be applied in order to achieve the desired stretch within the failure zone in order to allow for a use of the results within a multi scale framework.

Finally, the new homogenization method could be applied to investigate the in-situ strength effect of cross ply laminates as shown experimentally by [Parvizi et al. \(1978\)](#) and investigated numerically by [Guillén-Hernández et al. \(2020\)](#). However, new boundary conditions in accordance to those applied by [Guillén-Hernández et al. \(2020\)](#) would be needed.

CRediT authorship contribution statement

Lukas Poggenpohl: Adjusted the material model presented in previous article, Wrote the scripts for geometry generation, Set up the numerical simulations, Analyzed and Visualized the results, Wrote the main part of the publication. **Hagen Holthusen:** Gave valuable feedback during result discussion and paper writing. **Jaen-Willem Simon:** The work and paper writing was supervised, Gave valuable feedback during result discussion and paper writing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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